USER MANUAL

EnergyMonitor

Home electricity consumption monitoring system — web application and mobile application

Field	Detail
Project	EnergyMonitor – Electrical Consumption Monitor
Document type	User Manual (web and mobile, single document)
Team members	Juan Esteban Cortés Parra (Project lead / DB manager); Miguel Ángel García Artunduaga (Analyst / UX-UI)
Training program	Technologist in Software Analysis and Development (ADSO)
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Training center	SENA Industria – Neiva
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Version control

Version	Date	Description of the change	Owner
1.0	07/09/2026	Initial version of the EnergyMonitor user manual.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga
2.0	07/23/2026	Separate manuals for the web application and the mobile application, with homes, roles, consumption, history, devices, thresholds, favorites and notifications.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga
3.0	10/27/2026	Single manual for web and mobile, with the differences of each platform marked. Adds the administrator role and its web screens, the acceptance of the terms and the privacy policy, activity notifications, ownership transfer, changing the Wi-Fi network of a device, account deletion with a code for Google accounts and the changes of the mobile app (secure session, accessibility, page titles). Removes inviting members by email, which does not exist: joining a home is done with the access code.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga
3.1	10/28/2026	The Settings help center (documentation by role, support and news), member photos and the photo viewer, the unusual consumption alert and the Brave notifications notice are documented. Pending screenshots are marked.	
3.2	10/29/2026	Pending screenshots are replaced with the current captures of both applications (web: image-29 to image-41; mobile: jpeg files) and the references are synced across both languages.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga
3.3	10/30/2026	Clarifies that administration and the administrator role are exclusive to the web application: the mobile app has no administration notices and does not distinguish the role in Settings.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga

Version	Date	Description of the change	Owner
3.4	11/02 /2026	Annex A (images) is rewritten: it states where the screenshots are stored and their naming convention, confirms that every image referenced in the text is included, and removes the mention of the internal documentation update report.	Juan Esteban Cortés Parra, Miguel Ángel García Artunduaga

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Introduction

EnergyMonitor was born from a simple idea: that anyone can know how much they are spending on electricity without waiting for the utility to deliver the final bill. Behind the application there are measuring modules with a PZEM-004T V3 sensor and an ESP32 microcontroller, connected to the appliances of the home, which send every minute the information the system uses to build dashboards, charts, alerts and saving recommendations.

The system is used in two ways, with the same account and the same data: the **web application**, from any browser, and the **"EnergyMonitor" mobile application** for Android. This document explains both in one place: creating and verifying an account, signing in, recovering the password, creating homes or joining them, checking consumption and history, managing members, devices and limits, receiving notifications, adjusting preferences and, for administrator accounts, administering the platform from the web.

Objective

That any user, without help from the development team, can register, sign in and confidently use every screen available on the web and in the mobile app, and that an administrator knows how to manage the accounts from the web.

Scope

This manual covers everything an end user and a functional administrator do in EnergyMonitor: the public landing page (web only), registration with acceptance of the terms, email verification, sign-in (including Google), password recovery, the homes dashboard, creating homes and joining with the access code, the home detail with its six tabs, linking devices over Bluetooth, favorites, notifications (alerts, recommendations and activity), settings, the personal account and account administration (web only).

It does not cover the internal architecture, the source code or the database (technical manual), the installation of the system or the device wiring (installation manual), or the technical administration of the server (administrator manual).

Requirements to use the system

Requirement	Web application	Mobile application
Device	Computer or phone with an up-to-date browser (Chrome, Edge, Firefox or Safari).	Up-to-date Android phone (the app is not available for iPhone).
Connection	Stable internet.	Stable internet.
Account	A valid email address or a Google account.	The same account as on the web.
Linking devices	Chrome or Edge on a computer, or Chrome on Android, with Bluetooth and the web opened over HTTPS. Other browsers use the module's Wi-Fi portal.	Bluetooth on and the Bluetooth permission granted to the app.
Push notifications	Browser notification permission (on iPhone, install the web on the home screen).	Android notification permission.

The measuring module (ESP32 + PZEM-004T V3) is installed separately following the installation manual; here we only explain how to link it to a home.

How to read this manual

Each screen is described with the same structure: description, a table with the role that can access it, the access path and the components, steps to use it, expected result, notes and the reference image. Each section states on which platform it is available:

Label	Meaning
[Web and mobile]	The screen exists in both applications and works the same.
[Web only]	It exists only in the web application (for example, the landing page and the administration).
[Mobile only]	It exists only in the mobile application (for example, the phone's Bluetooth permission).
Web: / **Mobile:**	Inside a common section, marks a difference in presentation or steps.

Images carry the suffix *(web)* or *(mobile)* depending on the application they were taken from.

System roles

Role	Who it is	What they can do
Visitor	Person without a session.	See the landing page (web), register, sign in and recover the password.
User	Any verified account.	Create homes, join homes with a code, mark favorites, see their notifications, configure the app and manage their account.
Home owner (responsible)	Whoever created the home or received its ownership. Each home has only one.	Everything a member can do, plus: see and share the access code, link, edit and delete devices, change the thresholds, remove members, transfer ownership and delete the home.
Home member	Whoever joined with the access code.	View consumption, history, members, devices, thresholds and home information; receive its notifications; leave the home.
Administrator	Account with the global administrator role. **[Web only]**	View the account directory, change account status, grant or revoke the administrator role and review the administration notices. Sees no homes.

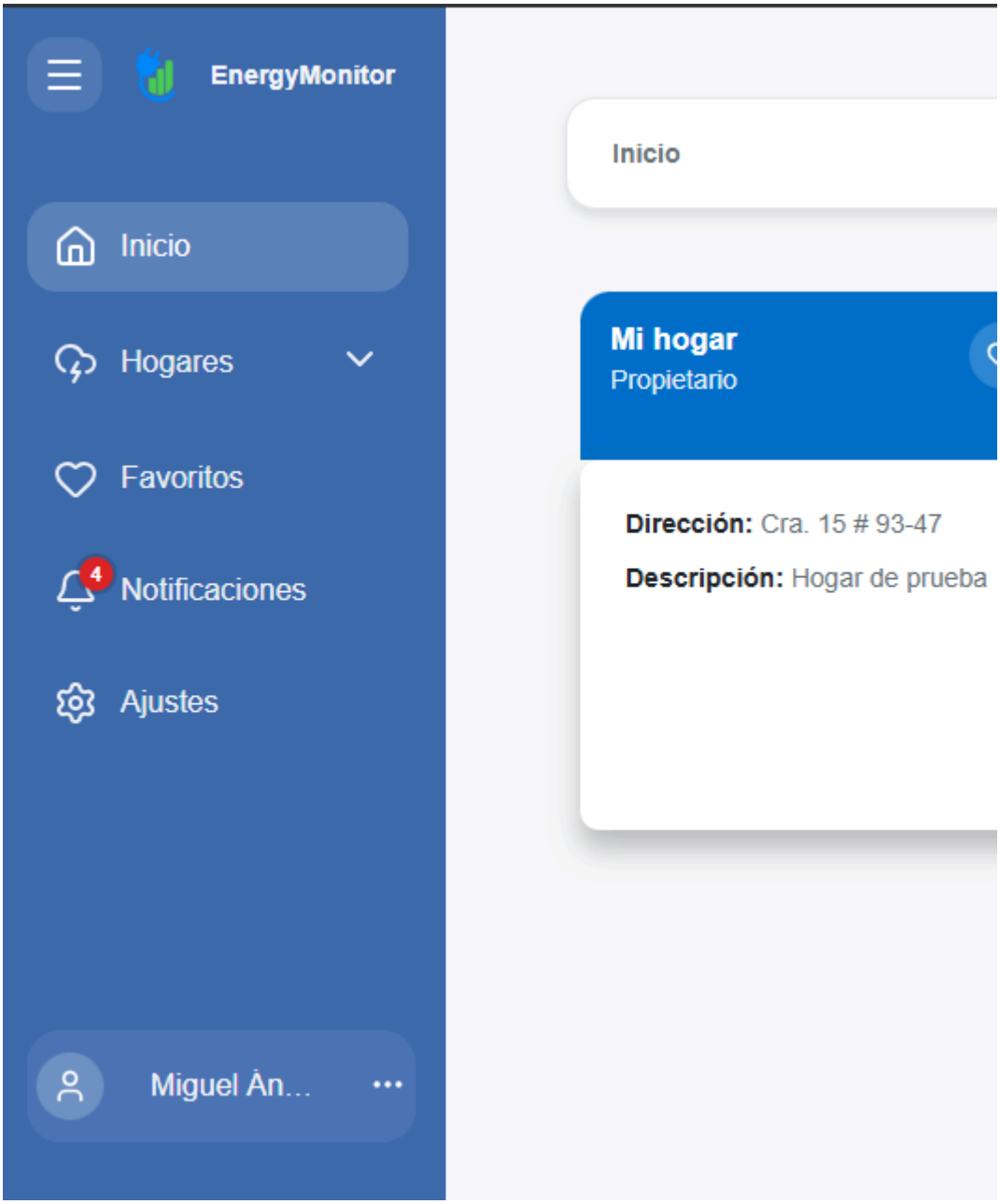
The same person can be the owner of one home and a member of another. The administrator role gives no access to any home.

General navigation

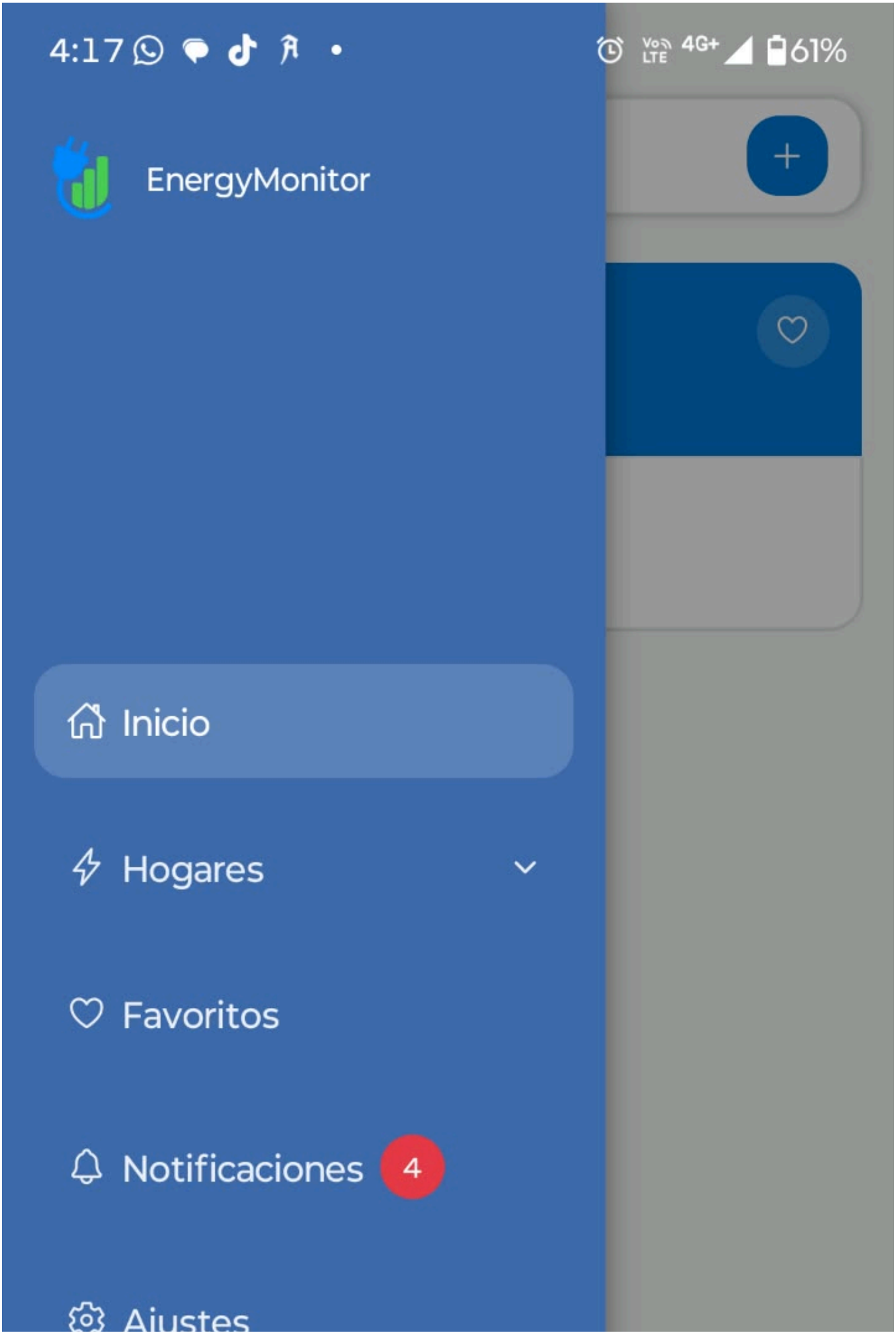
Web. After signing in, the left side menu shows **Home**, **Homes** (drop-down list of the user's homes), **Favorites**, **Notifications** (with the number pending) and **Settings**; at the bottom is the profile, with **My account** and **Sign out**. The menu can be collapsed; on small screens it opens with the menu button. An administrator account sees only **Administration**, **Notifications** and **Settings** (section 15).

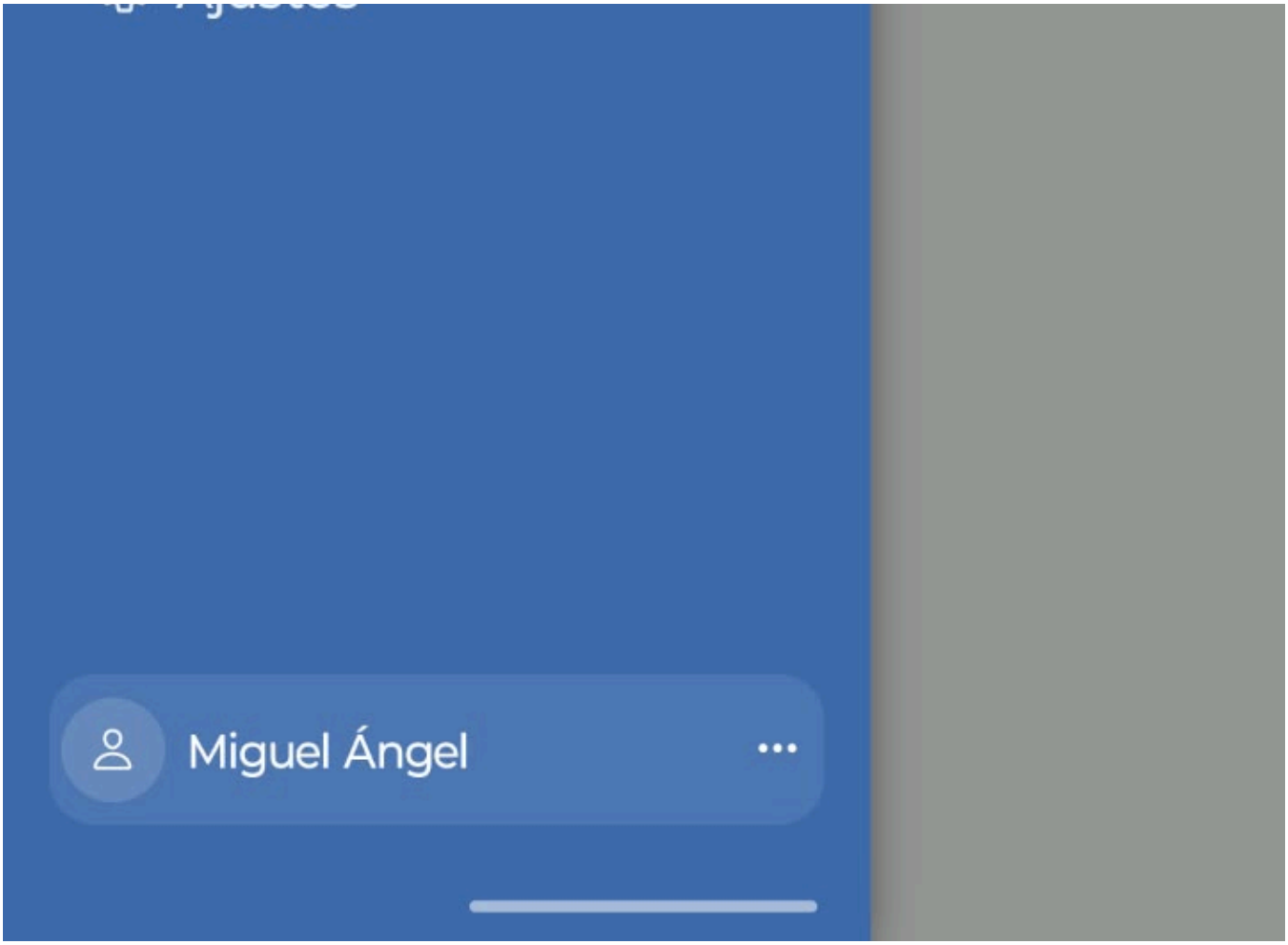
Mobile. The app opens directly on the sign-in screen (it has no landing page). After signing in, the side menu shows **Home**, the list of homes, **Favorites**, **Notifications** and **Settings**, and gives access to **My account**; Android's back button closes the menu. Inside a home, the six tabs are reached from the bottom bar. The session is stored securely on the phone and kept between app launches.

Both applications are available in **Spanish, English, French and Portuguese**, have **four themes** (two light and two dark) and work with screen readers.



Side menu (web)


4:17
61%
EnergyMonitor
Inicio
Hogares
Favoritos
Notificaciones
4
Ajustes



Side menu (mobile)

1. Landing page (web only)

[Web only] The mobile app has no landing page: it opens directly on the sign-in screen.

Description: The entry door to the system: the public screen anyone can see without an account. It presents what EnergyMonitor is about and gives access to Sign in or Register.

Field	Detail
Role that can access	Visitor (no account needed).
Access path	Main URL of the system (`/home`).
Screen components	Navigation bar with the Home, About us and Support links; language selector; button to switch to light or dark mode; Sign in and Register buttons; welcome, How it works and Benefits sections; footer.

Steps to use it:

1. Open the EnergyMonitor URL in the browser.
2. Scroll the page to learn what the system does (How it works and Benefits sections).
3. For more context about the project, open About us; for questions, Support.
4. Click Register for the first time, or Sign in if you already have an account.

Expected result: The person understands what the platform is for and continues to registration or sign-in.

Notes: The language and theme chosen here are kept inside the application.



Landing page (web)

How it works presents the general flow, from registering to receiving smart alerts.



How it works (web)

Benefits summarizes why the system is worth using: saving on the bill by finding unnecessary consumption, alerts on peaks or unusual activity, a smaller environmental footprint and a simple setup.



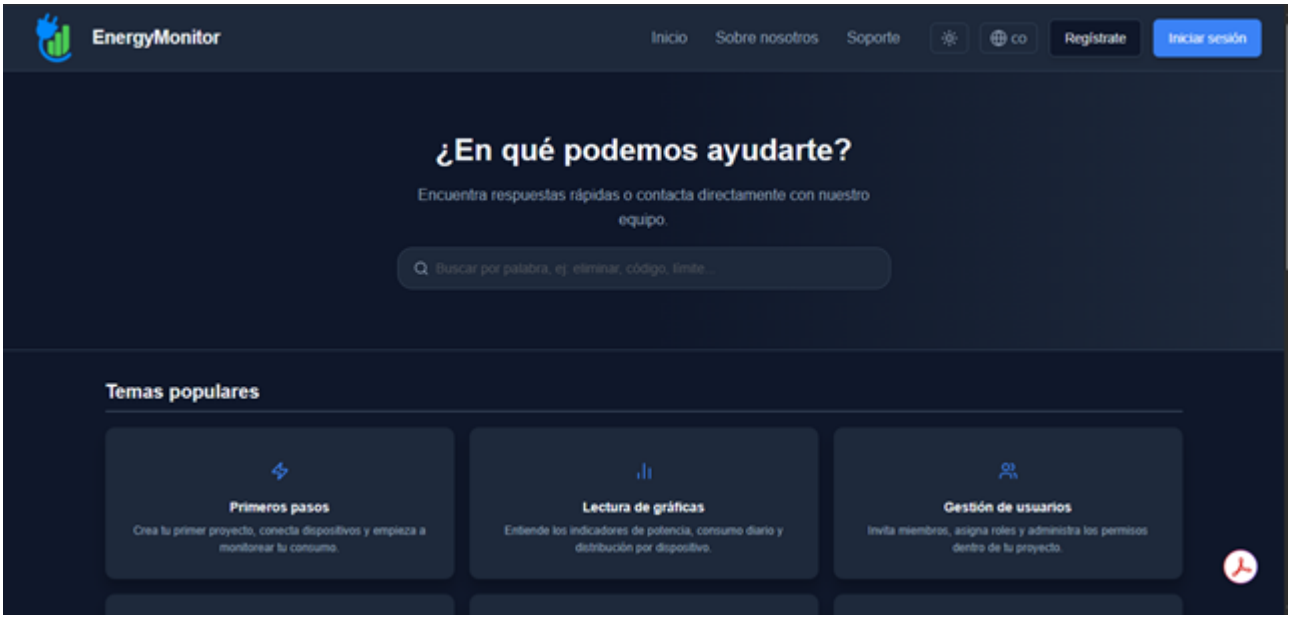
Benefits (web)

About us tells the problem EnergyMonitor seeks to solve, the mission of the project and the hardware that makes monitoring possible.



About us (web)

Support gathers the popular topics, a search box for frequently asked questions (for example "delete", "code" or "limit") and the contact with the team: the page shows the active administrators of the platform so you can write to them by email; when there are several, you choose one.



Support (web)

2. Registration

[Web and mobile]

Description: Form where a new person creates their account with first name, last name, email and password, or with their Google account. The account is only created when the email is verified (section 3).

Field	Detail
Role that can access	Visitor.
Access path	**Web:** Landing page > Register (`/register`). **Mobile:** sign-in screen > Register.
Screen components	First name, Last name (optional), Email, Password and Repeat password fields; persistent hint with the password rules; checkbox "I agree to the Terms of use and the Privacy policy", whose links open the texts in a window; Register button; Register with Google button; Already have an account? Sign in link.

Steps to use it:

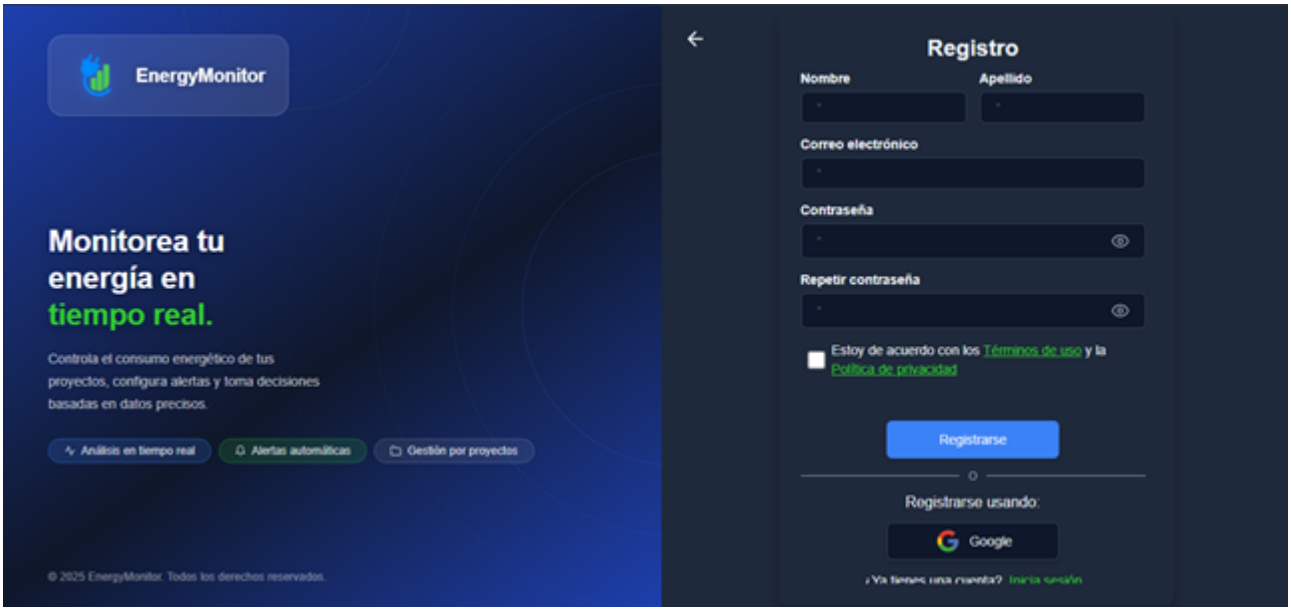
> 1. Open the registration.
> 2. Fill in first name, last name if you wish, email and the password twice. The password must have between 8 and 64 characters, with at least one upper-case letter, one digit and one special character; the rule is always shown under the field.
> 3. To read the conditions, tap **Terms of use** or **Privacy policy**: a window opens with both tabs (the same texts on the web and in the app). Close it and tick the acceptance checkbox.
> 4. Press **Register**. To use Google, press **Register with Google** with the checkbox ticked.

Expected result: The system sends a 6-digit code to the email and opens the verification screen. With Google, the account is created at once, already verified.

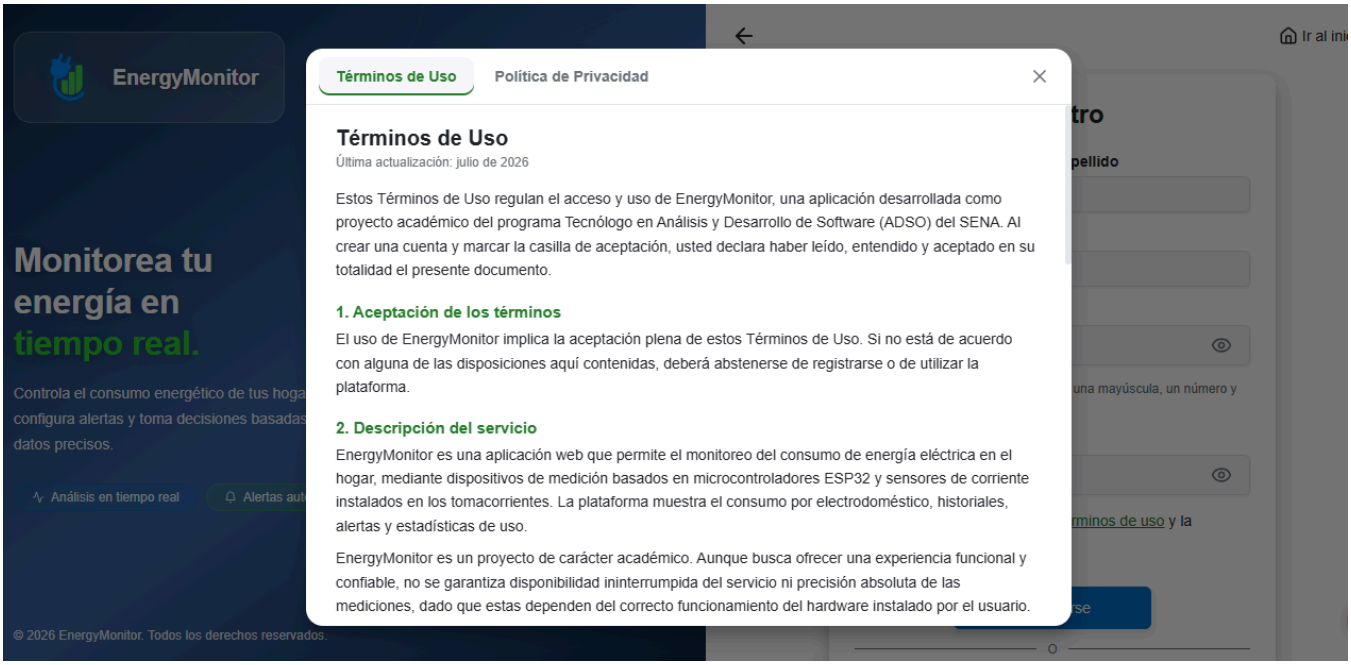
Notes:

- Without accepting the terms the account cannot be created, neither with email nor with Google. When a new Google account tries to enter without having accepted them, the application shows the terms and, once accepted, opens Google again (web) or repeats the sign-in (mobile).
- If the email already has an account, the system says so and suggests signing in.

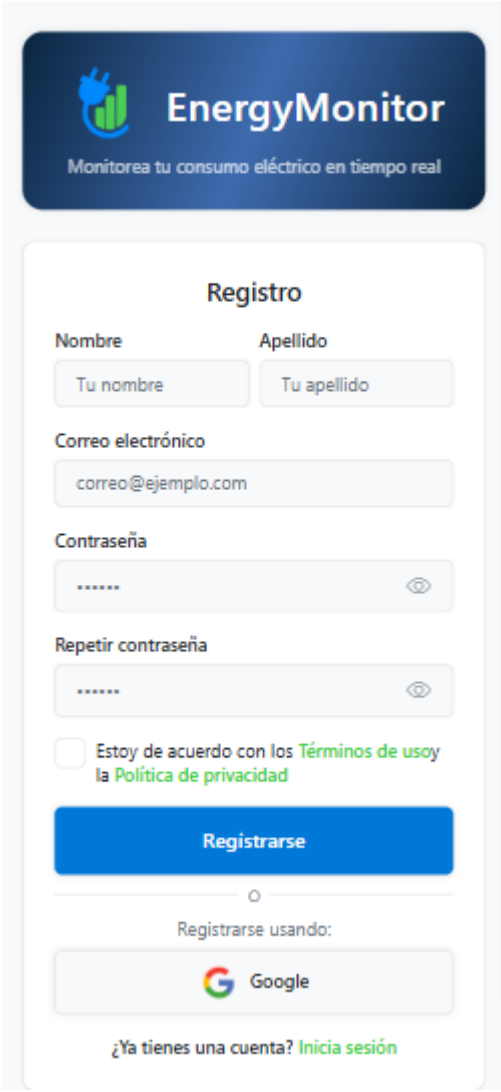
- If the email domain looks mistyped (for example `gmial.con`), the system refuses it and suggests the right one; a domain that does not exist is also refused.
- Registering the same email again before verifying it replaces the previous request and its code stops working.



Registration (web)

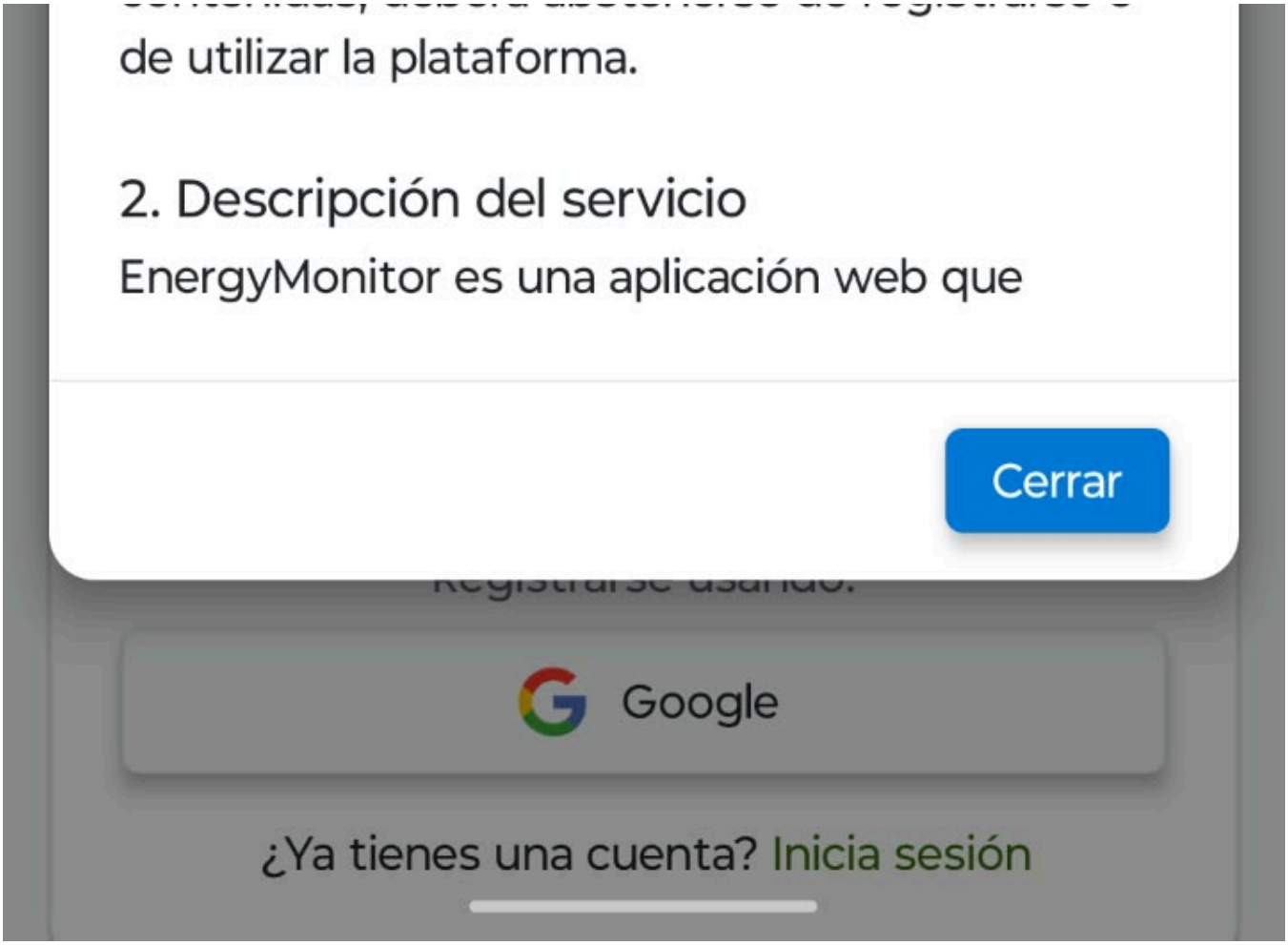


Terms of use and Privacy policy window (web)



Registration (mobile)


2:52
66%
EnergyMonitor
Documentos legales
Términos de Uso
Política de Privacidad
Términos de Uso
Última actualización: julio de 2026
Estos Términos de Uso regulan el acceso y uso de EnergyMonitor, una aplicación desarrollada como proyecto académico del programa Tecnólogo en Análisis y Desarrollo de Software (ADSO) del SENA. Al crear una cuenta y marcar la casilla de aceptación, usted declara haber leído, entendido y aceptado en su totalidad el presente documento.
1. Aceptación de los términos
El uso de EnergyMonitor implica la aceptación plena de estos Términos de Uso. Si no está de acuerdo con alguna de las disposiciones aquí



Terms of use and Privacy policy window (mobile)

3. Account verification

[Web and mobile]

Description: Screen that confirms the email used at registration is real. Typing the 6-digit code creates the account.

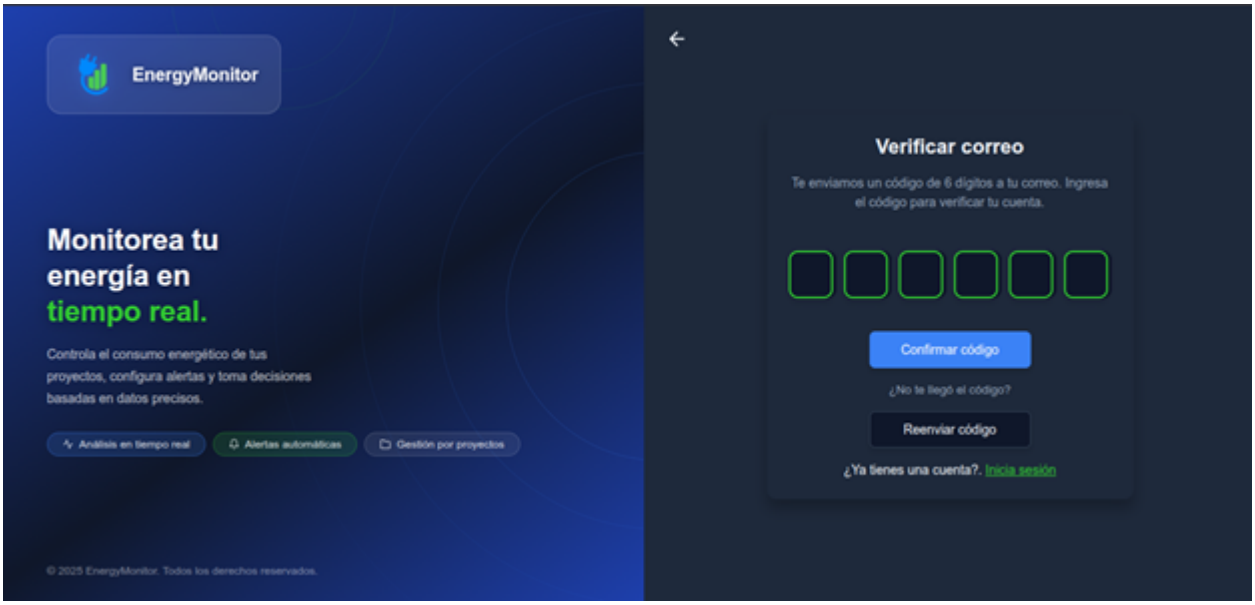
Field	Detail
Role that can access	Person who has just registered (or who signed in without having verified their email).
Access path	Opens automatically at the end of the registration.
Screen components	Six boxes for the code; Didn't get the code? link to send it again (with a wait between sends); link to Sign in.

Steps to use it:

1. Check the inbox (and the spam folder) of the registered email.
2. Type the 6-digit code. The whole code can also be pasted at once.
3. Confirm.

Expected result: The account is created and verified; the system shows "Email verified. You can now sign in.".

Notes: The code is valid for 15 to 30 minutes and allows few failed attempts. When it expires, ask for a new one with Didn't get the code?. A registration request not verified within 24 hours is deleted and you must register again.



Account verification (web)



Account verification (mobile)

4. Sign in

[Web and mobile]

Description: Form to enter with a verified account, with email and password or with Google.

Field	Detail
Role that can access	Visitor with an account.
Access path	**Web:** Landing page > Sign in (`/login`). **Mobile:** first screen of the app.
Screen components	Email and Password fields (with a button to show or hide it); Forgot your password? link; Sign in button; Google button; Don't have an account? Register link.

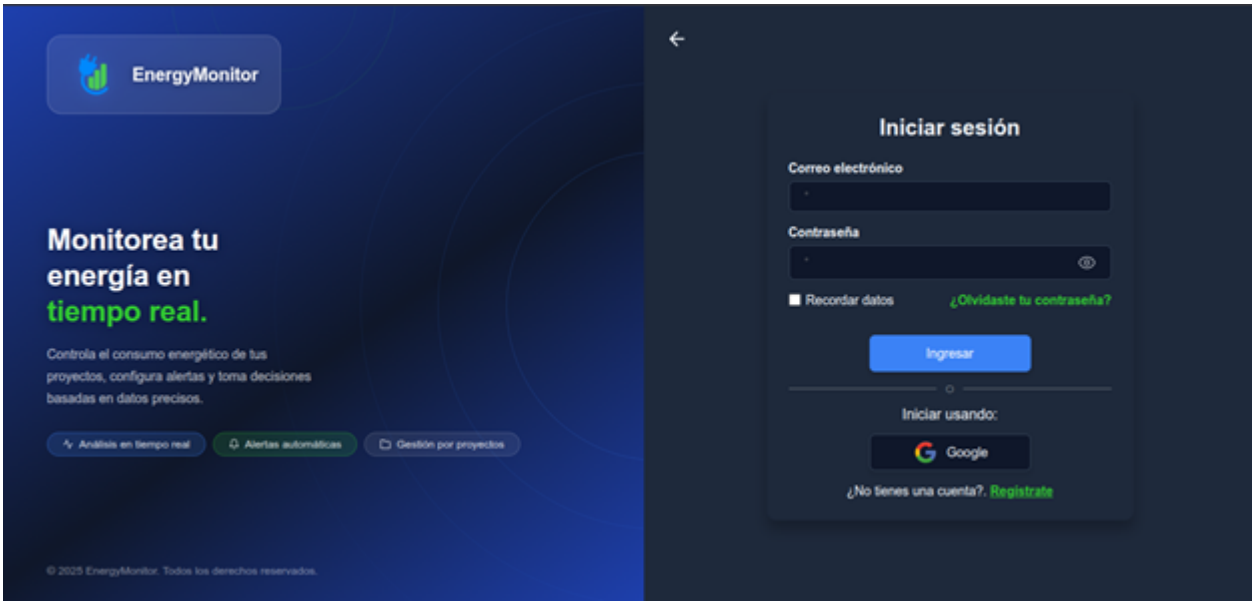
Steps to use it:

1. Type the email and the password, or press the Google button.
2. Press Sign in.

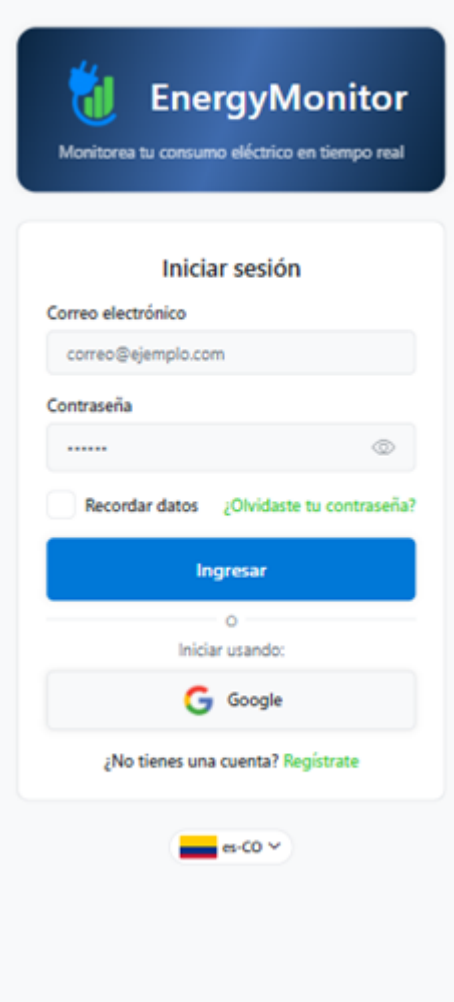
Expected result: The homes dashboard opens (or the administration, if the account is an administrator and the web is used).

Notes:

- If the email is not verified yet, the system says so, sends a new code and takes you to the verification.
- If the account has not accepted the terms and conditions yet, the system asks for them before entering.
- An account deactivated or blocked by an administrator cannot sign in.
- On the web, with a session open, opening the login or the registration goes straight to the dashboard.



Sign in (web)



Sign in (mobile)

5. Password recovery

[Web and mobile]

Description: Lets you set a new password when you forgot the current one, with a 6-digit code sent by email.

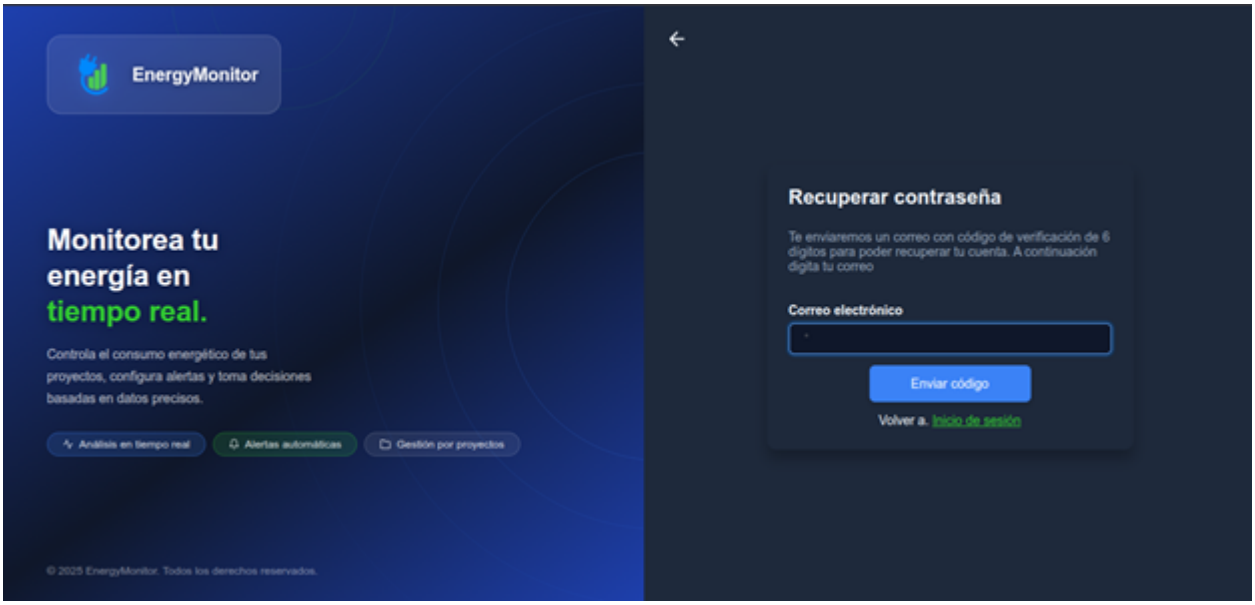
Field	Detail
Role that can access	Visitor with an account.
Access path	Sign in > Forgot your password?
Screen components	Step 1: Email field and button to send the code. Step 2: code boxes. Step 3: New password and Repeat password, with the rules hint. **Mobile:** step 3 is a screen of its own ("New password").

Steps to use it:

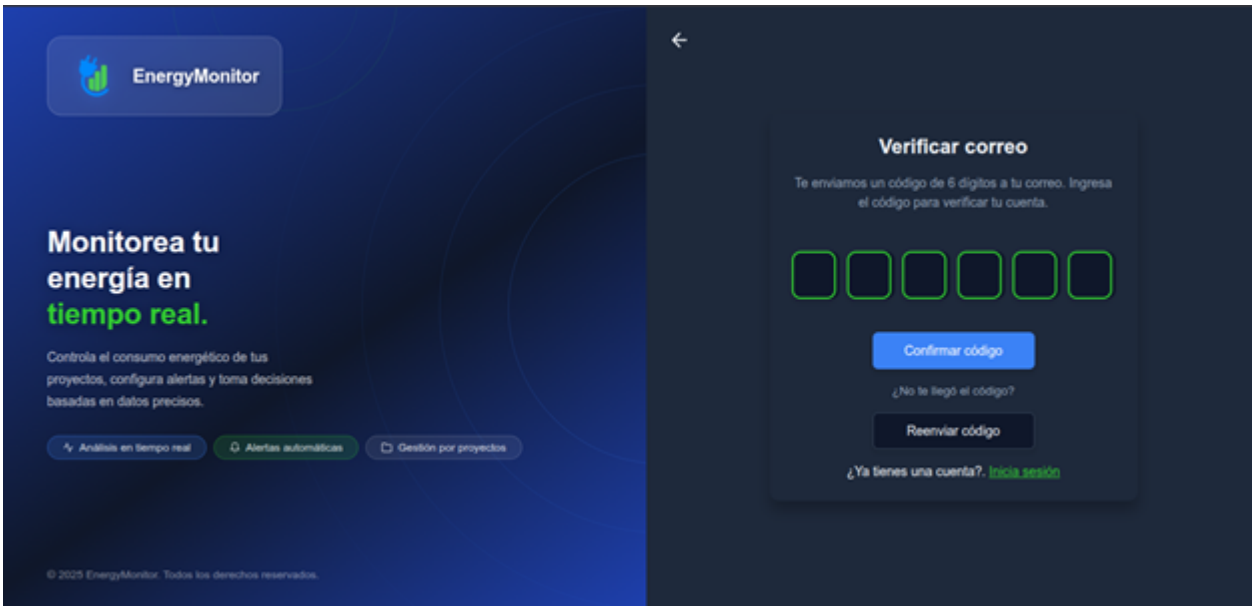
> 1. Type the email of the account and send.
> 2. Type the received code (valid for 15 minutes and usable once).
> 3. Type the new password twice and confirm.

Expected result: The password is updated ("Password updated. Sign in with the new one.").

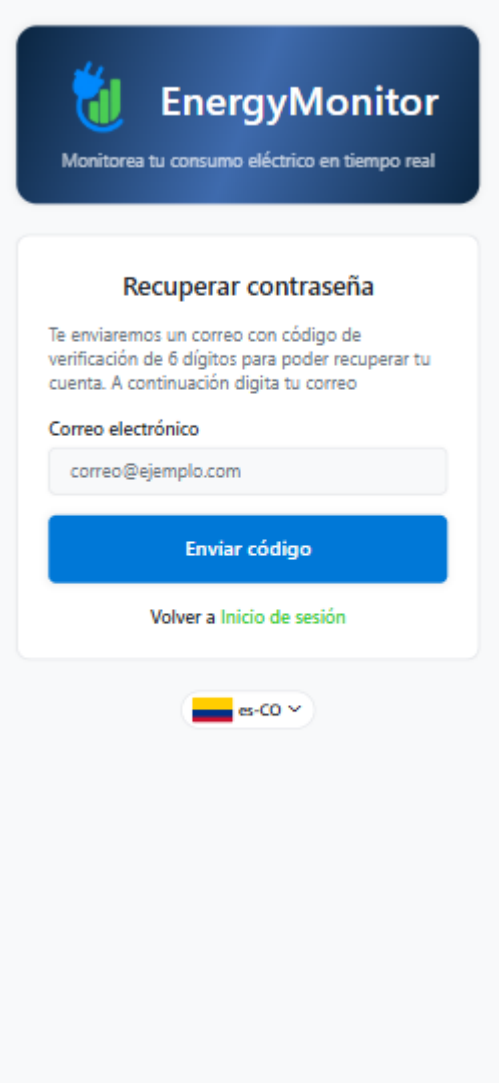
Notes: If the email has no account, the system says so. Asking for a new code invalidates the previous one. After several failed attempts you must wait a few minutes. Accounts created with Google can use this option to also have a password.



Password recovery (web)



Recovery verification (web)



Password recovery (mobile)



Recovery verification (mobile)

6. Homes dashboard

[Web and mobile]

Description: The main screen after signing in. It shows the cards of the user's homes, or an invitation to create or join the first one.

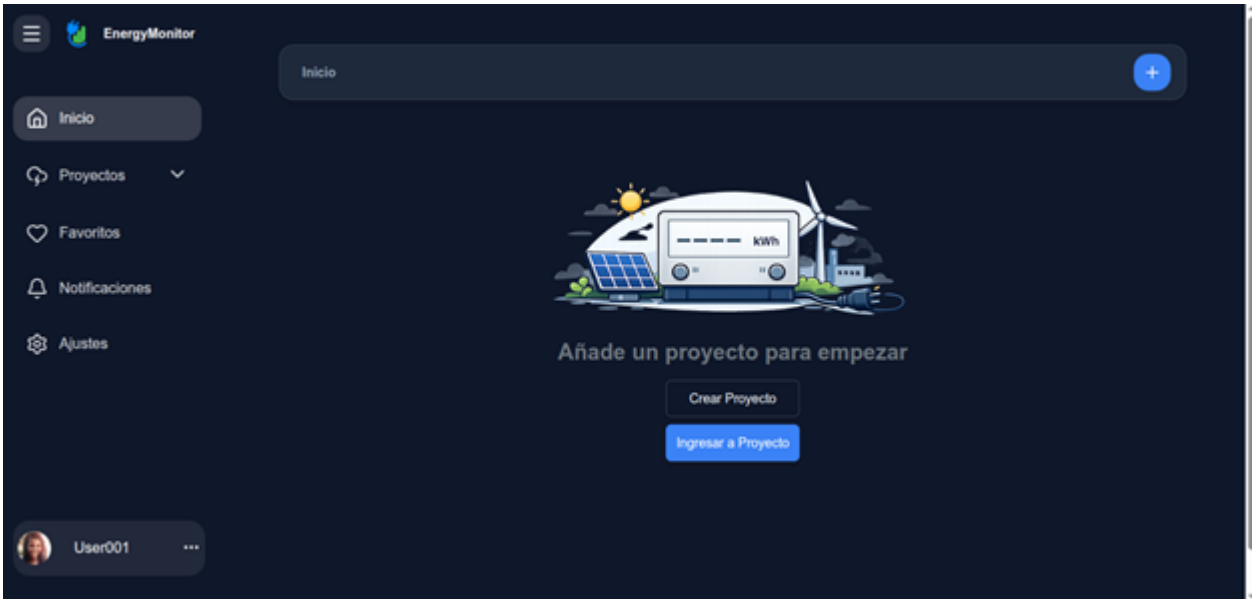
Field	Detail
Role that can access	User (not administrator).
Access path	Opens after signing in; also from **Home** in the side menu.
Screen components	Home cards with the name, the type, the user's role (Owner or Member) and the heart button to mark it as favorite; **+** button to Create home or Join home; empty state with the same buttons when there are no homes.

Steps to use it:

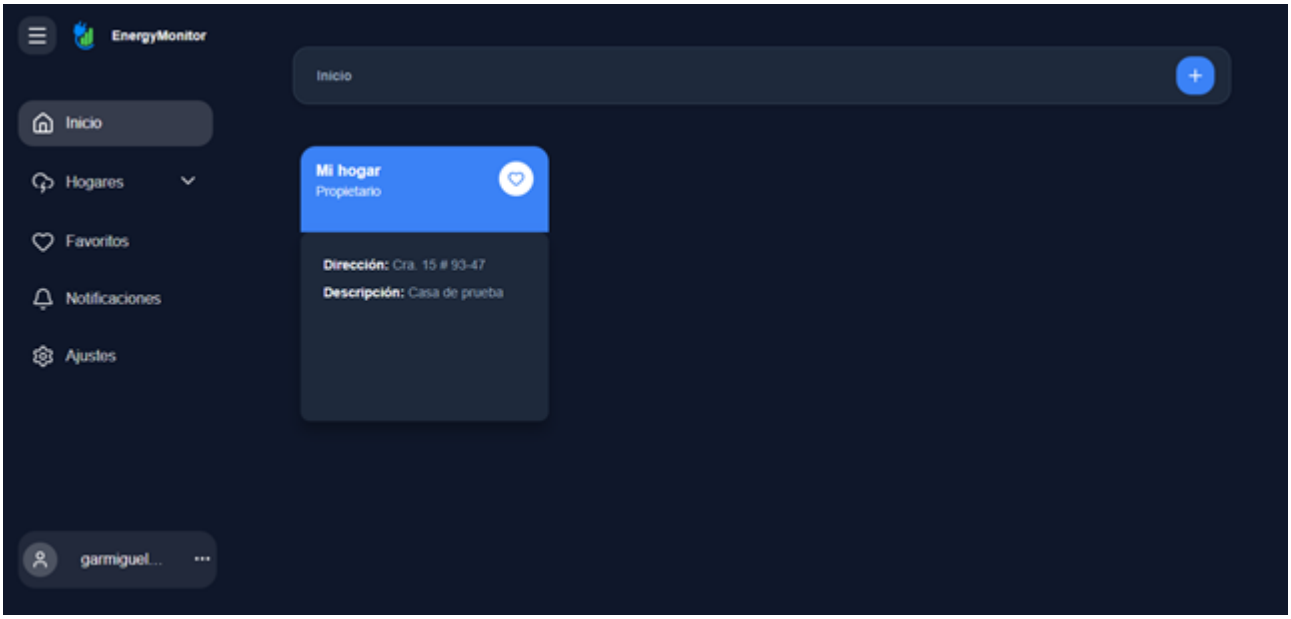
> 1. Sign in: the dashboard appears by itself.
> 2. If there are no homes, choose Create home (section 7) or Join home (section 8).
> 3. Tap a home card to open its detail (section 9).

Expected result: The user sees their homes at a glance and enters the one they need.

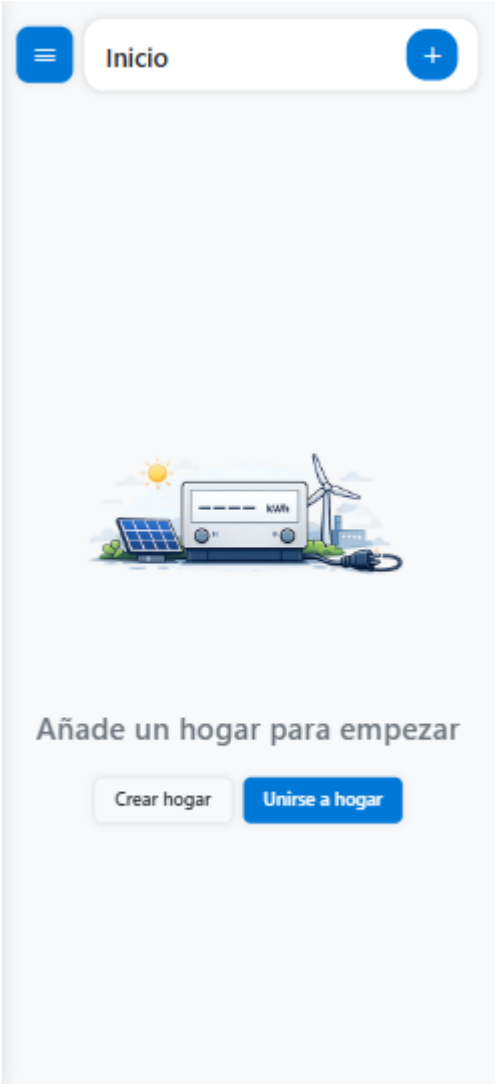
Notes: Changes made by other members (for example, a deleted home or a transferred ownership) show when you come back to the dashboard.



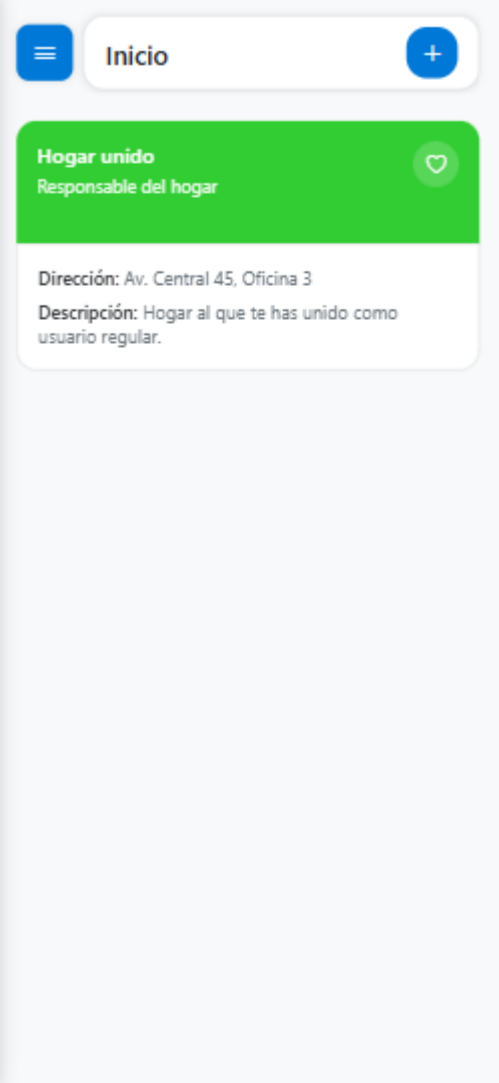
Dashboard without homes (web)



Dashboard with homes (web)



Dashboard without homes (mobile)



Dashboard with homes (mobile)

7. Create a home

[Web and mobile]

Description: Form to register a new home. Whoever creates it becomes its owner.

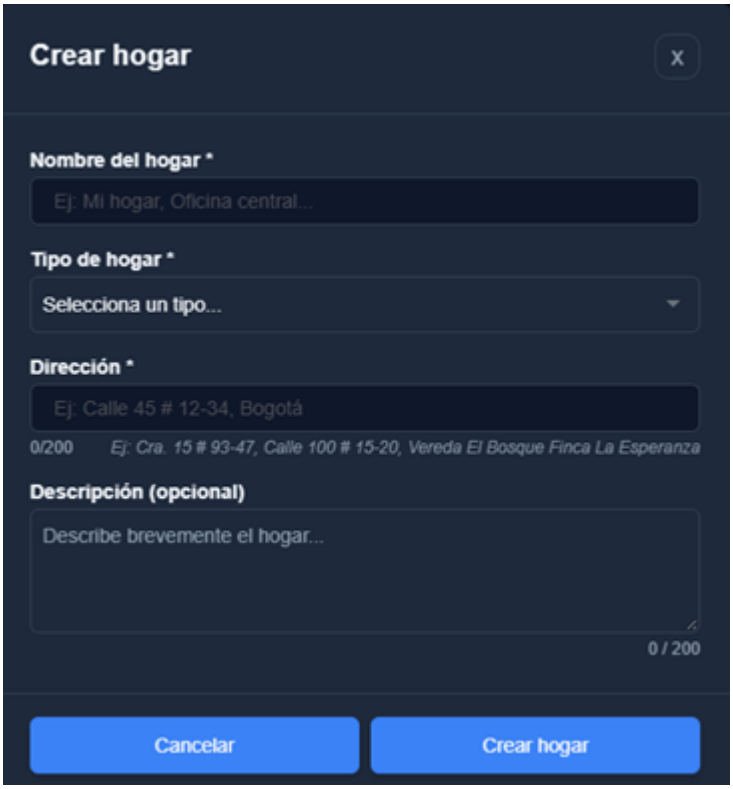
Field	Detail
Role that can access	User.
Access path	Homes dashboard > **+** button > Create home, or the button of the empty state.
Screen components	Home name (up to 50 characters), Home type (House, Apartment, Studio apartment, Country house, Cabin or Other; with Other it asks "Which one?"), Address (up to 200 characters, for example "Cra. 15 # 93-47") and Description (optional) fields; Cancel and Create home buttons.

Steps to use it:

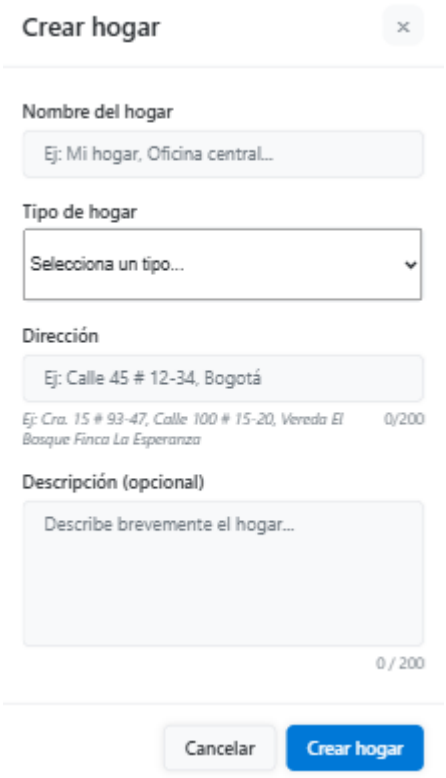
> 1. Open Create home.
> 2. Fill in name, type and address; the description is optional.
> 3. Press Create home.

Expected result: The card of the new home appears with the Owner role and the system gives it an 8-character access code.

Notes: The address does not accept the characters `< > { } □ | "` '`. The home starts with the system's default thresholds (section 9.5).



Create home (web)



Create home (mobile)

8. Join a home

[Web and mobile]

Description: Lets you enter an existing home as a member with the access code its owner shares. **It is the only way to add people to a home:** the system sends no email invitations.

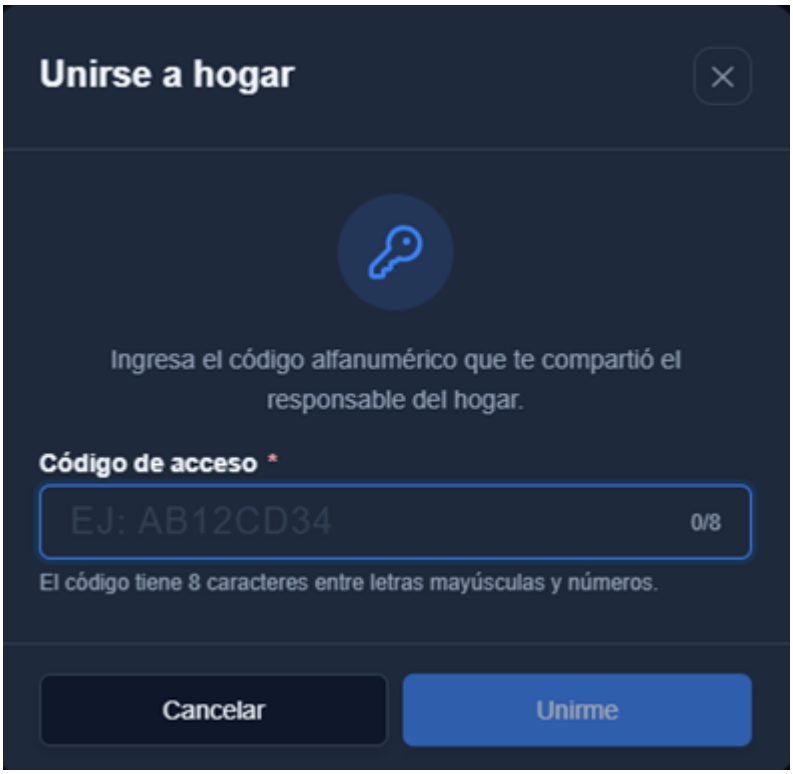
Field	Detail
Role that can access	User.
Access path	Homes dashboard > **+** button > Join home, or the button of the empty state.
Screen components	Access code field (8 characters of upper-case letters and digits, for example `AB12CD34`); Cancel and Join buttons.

Steps to use it:

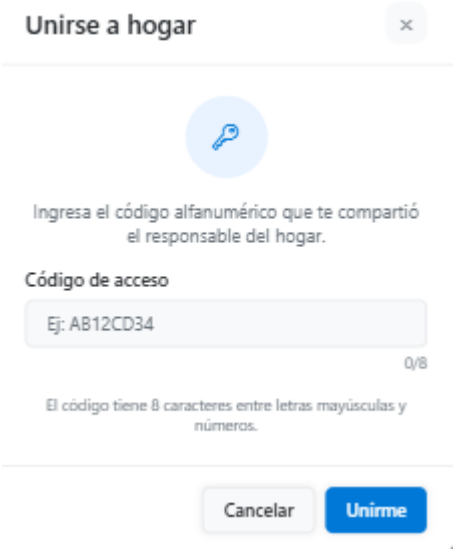
1. Ask the owner of the home for the access code (they see it in the Home tab, section 9.6).
2. Open Join home, type the code and press Join.

Expected result: The home card appears with the Member role. The other members receive an activity notification.

Notes: A member only views the information of the home; the management actions belong to the owner.



Join a home (web)



Join a home (mobile)

9. Home detail

[Web and mobile] Tapping a home opens its detail with six tabs: **Consumption**, **History**, **Users**, **Devices**, **Thresholds** and **Home**. **Web:** the tabs are at the top and can be reached with the keyboard. **Mobile:** they are on the bottom bar.

9.1 Consumption

Description: Tab opened by default. It shows consumption almost in real time.

Field	Detail
Role that can access	Owner and members of the home.
Access path	Dashboard > home card > Consumption.
Screen components	Indicators of current power, today's consumption, devices (all operational or how many disconnected) and consumption level (Low, Medium, High or Critical); current limit; chart of the last 24 hours (active power in kW); current distribution and consumption per device.

Steps to use it:

1. Open the home: the Consumption tab is shown.
2. Check the indicators and scroll to see the charts.

Expected result: The user understands how the consumption of the home is going right now.

Notes: The data depends on the linked devices; a disconnected one adds no readings. If the home has no devices, the screen says so. **Mobile:** the indicators use icons.



Consumption (web)



Consumption (mobile)

9.2 History

Description: Shows the consumption of the home over time.

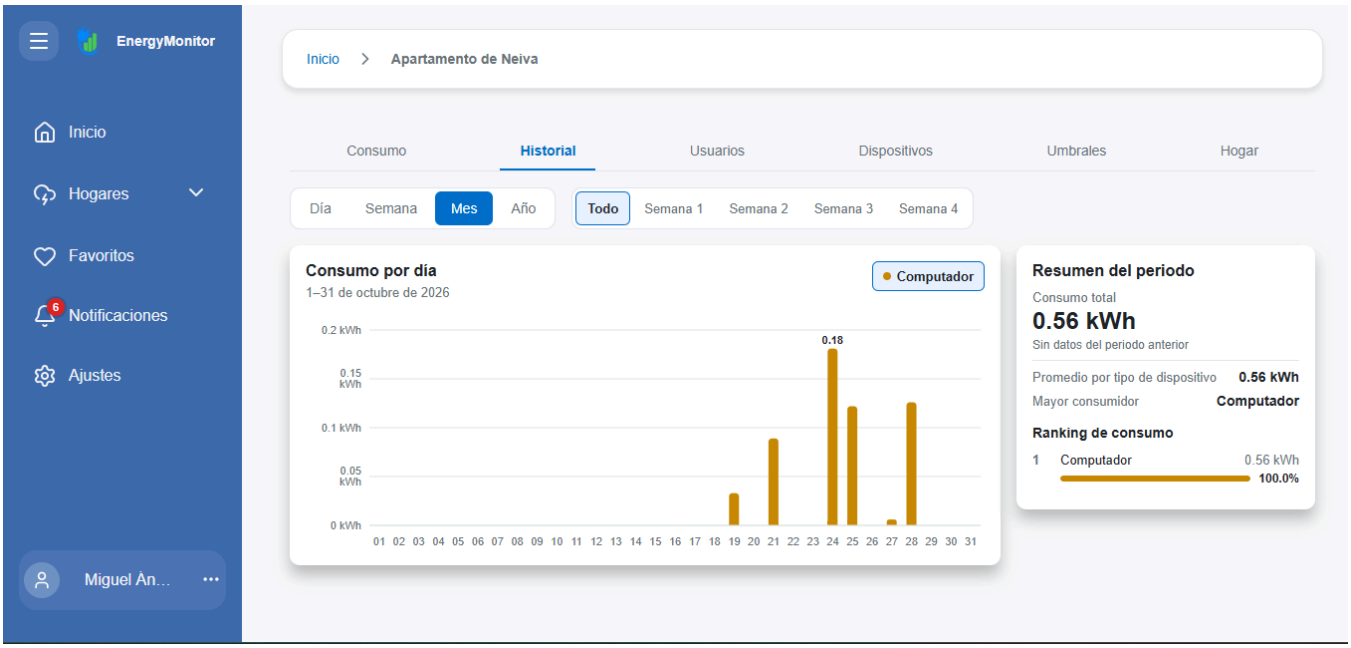
Field	Detail
Role that can access	Owner and members of the home.
Access path	Home detail > History.
Screen components	Period selector (Day, Week, Month, Year); interval inside the period (6-hour slots in Day, weeks in Month, quarters in Year, or All); chart of consumption per hour, day or month by device type; period summary (total consumption, top consumer and change against the previous period); consumption ranking per device.

Steps to use it:

1. Open the History tab.
2. Choose the period and, if you wish, an interval.
3. Check the chart, the summary and the ranking.

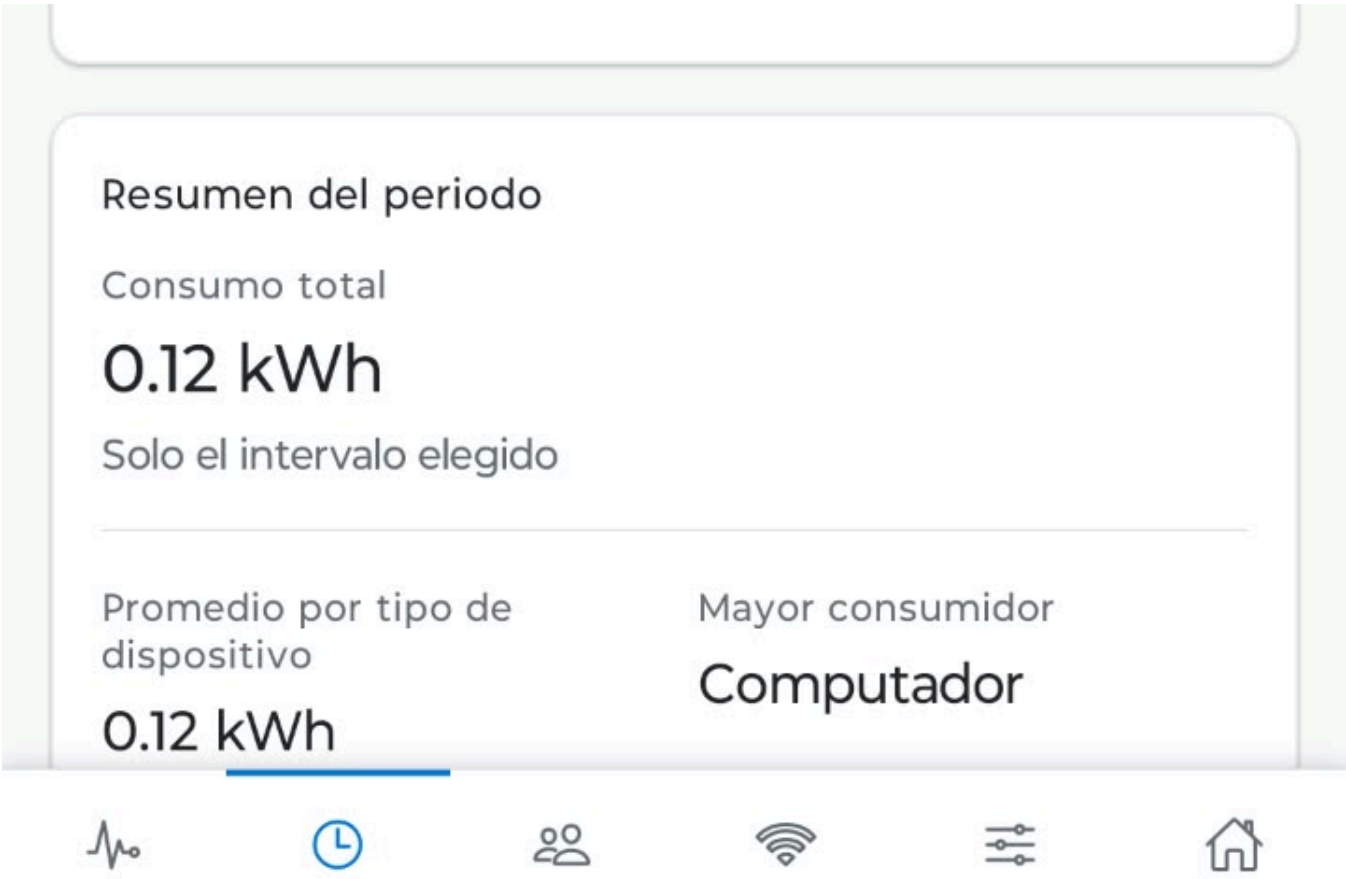
Expected result: The user compares periods and finds which appliance consumes the most.

Notes: The mobile app lays out the history the same way as the web. The trend arrow against the previous period is shown as an icon.



History (web)


Apartamento de Neiva
Día
Semana
Mes
Año
Todo
Semana 1
Semana 2
Semana 3
Consumo por día
1 de octubre de 2026 – 31 de octubre de 2026
Computador
0.09
0.07
0.04
0.02
0.00
15
16
17
18
19
20
21



History (mobile)

9.3 Users

Description: Lists who belongs to the home and their role.

Field	Detail
Role that can access	Owner and members (actions only for the owner).
Access path	Home detail > Users.
Screen components	Number of members; list with name, email and role (Owner or Member), with the "You" mark on your own account. **Owner:** buttons to remove a member and to make them owner.

Steps to use it (owner):

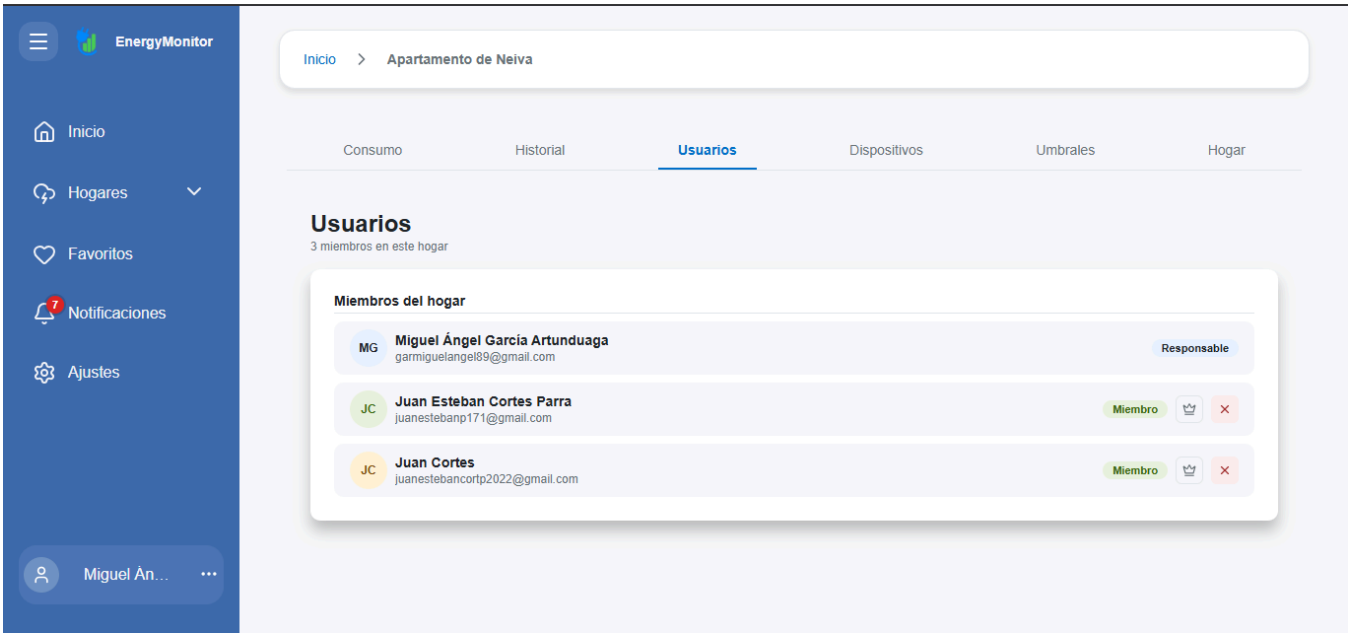
> 1. Open the Users tab.
> 2. **Remove a member:** press the delete button on their row and confirm. They lose access and must join again with the code.
> 3. **Transfer ownership:** press "Make owner" on the member's row, tick "I understand I will stop being the owner" and press Transfer.

Steps to use it (member): open the tab and view the list.

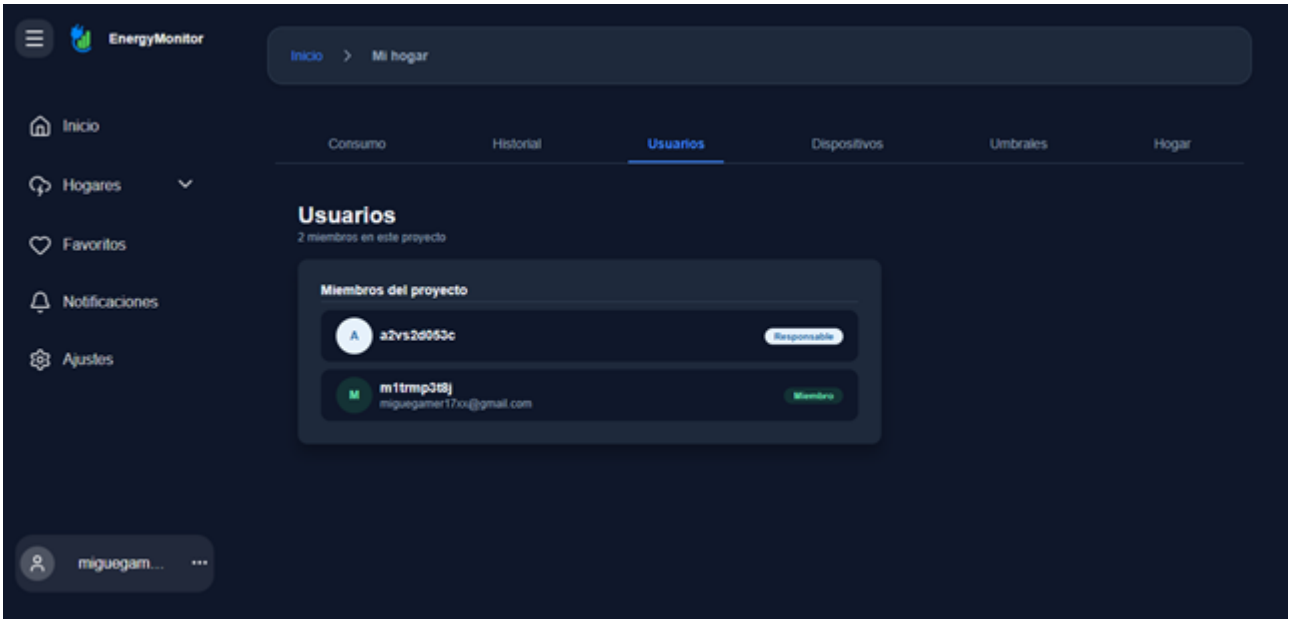
Expected result: The owner keeps the list of members up to date; after a transfer, the new owner manages the home and whoever transferred it stays as a member.

Notes: If the home only has its owner, the screen invites you to share the access code. Devices, thresholds and history do not change with the transfer. Whoever is removed and the other members receive an activity

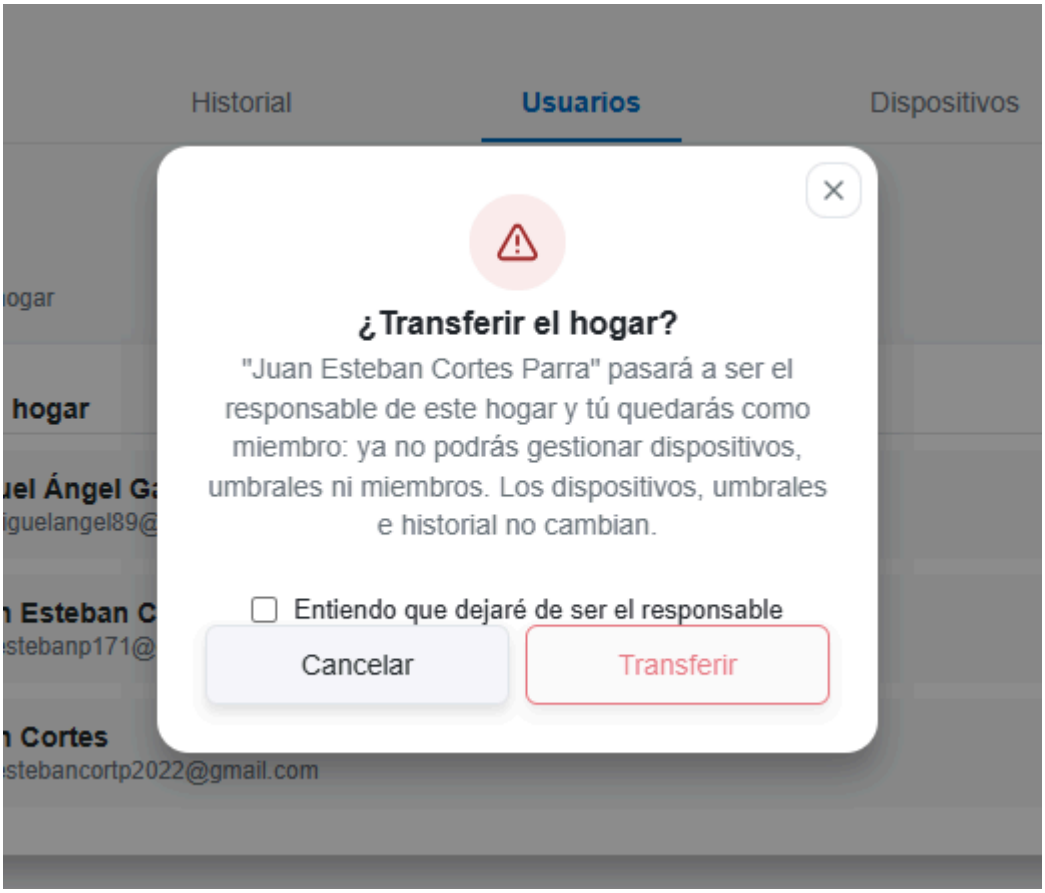
notification. **Web:** each member is shown with their profile photo (if they have one) and clicking it opens it in a viewer inside the page, with initials as fallback. **Mobile:** only initials are shown for now.



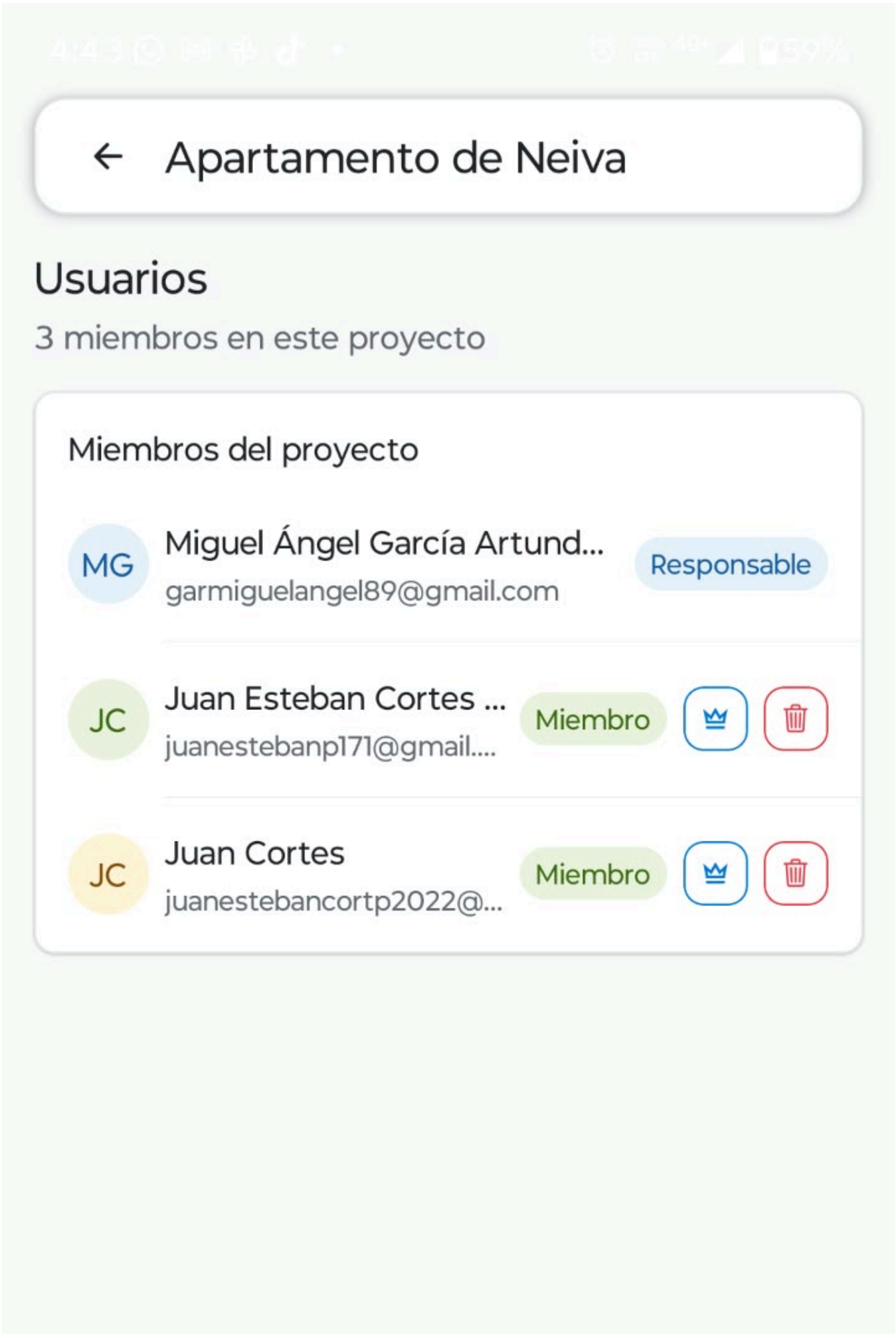
Users, owner view (web)

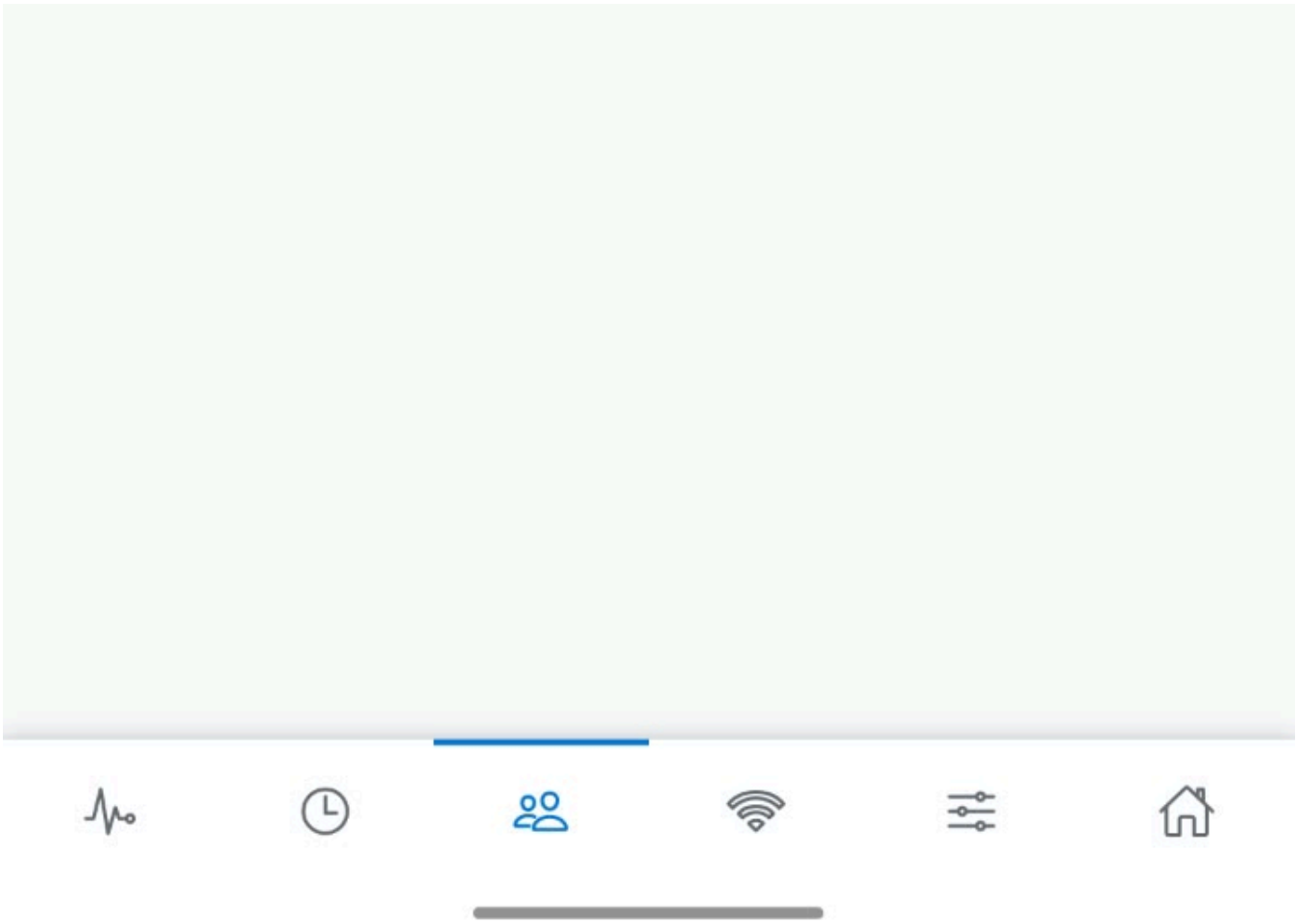


Users, member view (web)

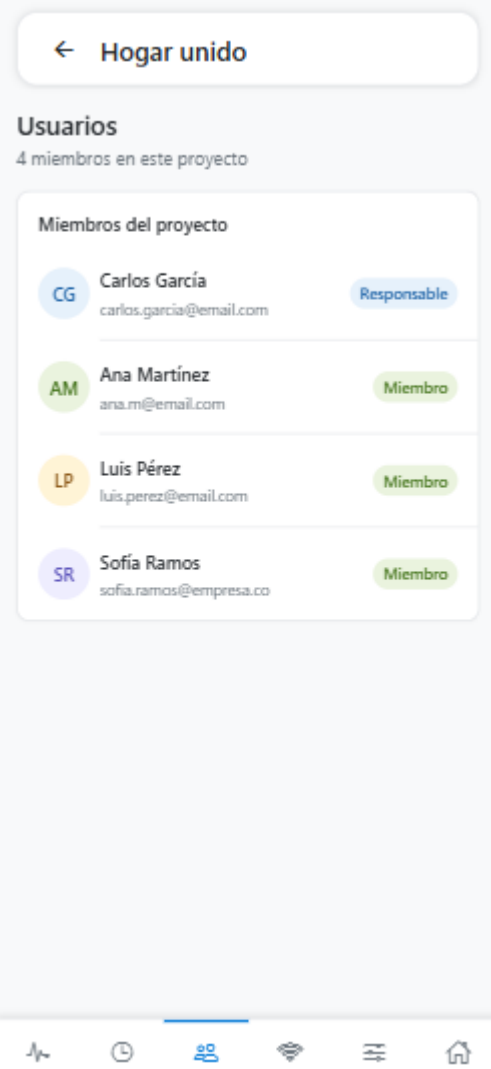


Ownership transfer confirmation (web)


Apartamento de Neiva
Usuarios
3 miembros en este proyecto
Miembros del proyecto
MG
Miguel Ángel García Artund...
garmiguelangel89@gmail.com
Responsable
JC
Juan Esteban Cortes ...
juanestebanp171@gmail....
Miembro
JC
Juan Cortes
juanestebancortp2022@...
Miembro



Users, owner view (mobile)



Users, member view (mobile)

9.4 Devices

Description: Lists the measuring modules linked to the home with their status, and lets the owner link, edit, change the Wi-Fi network of or delete a module.

Field	Detail
Role that can access	Owner and members (actions only for the owner).
Access path	Home detail > Devices.
Screen components	Header "X of Y devices online"; cards with the name, the appliance, the location and the status (Online, Disconnected or Checking…). **Owner:** Link device button and, on each card, Edit, Change the Wi-Fi network and Delete.

Link a device (owner):

> 1. Plug in the module and hold its **BOOT** button for 3 seconds, until the light blinks (a new module is already ready).
>
> 2. Press **Link device** and then **Search module**. **Web:** the browser opens its Bluetooth window; choose `EnergyMonitor-<code>`. **Mobile:** the app asks for the Bluetooth permission the first time, looks for the nearby modules and shows them in a list; choose the module.
>
> 3. Choose which appliance it will measure (Fridge, Washer, TV, Microwave, Air conditioner, Computer, Water heater, Lighting or Other).

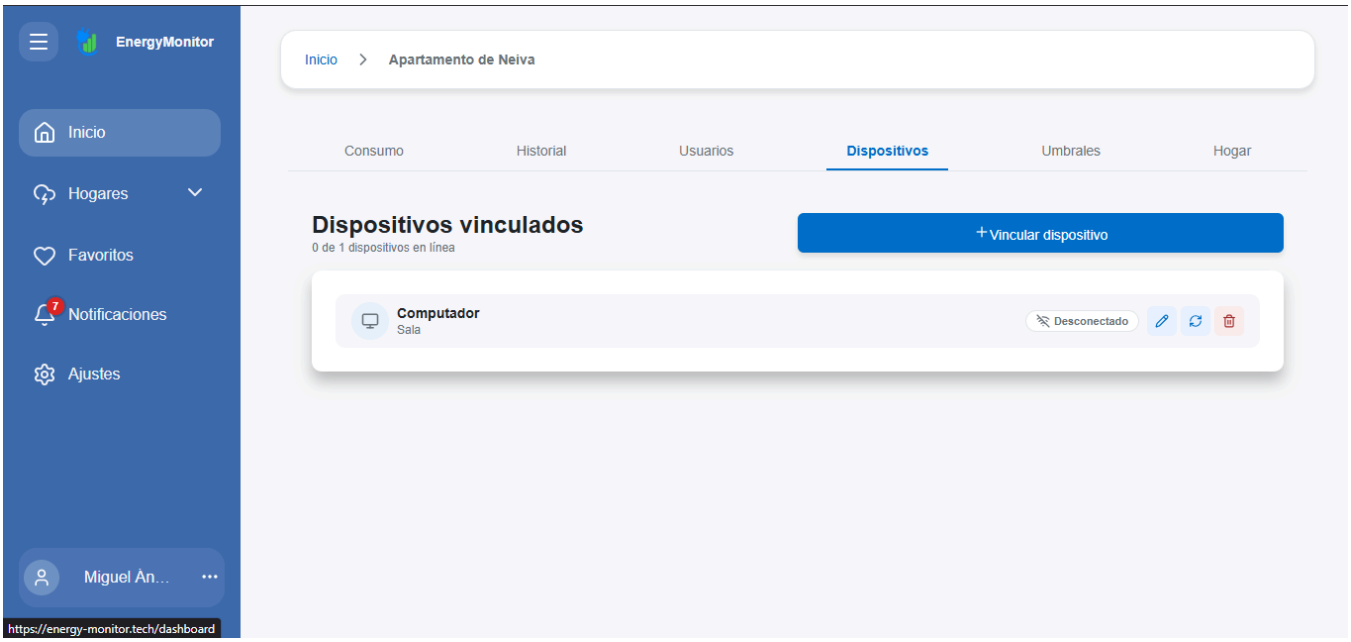
> 4. Choose the home's Wi-Fi network among those the module sees (or type its name by hand) and its password; press Connect.
>
> 5. Wait for the steps "Registering the device in your home", "Connecting the module to the network", "Checking the password" and "Connecting to the server".
>
> 6. On "Device linked!", type its name and location (Living room, Kitchen, Bedroom, Garage, Laundry room or Other) and press Finish.

Other actions (owner): Edit changes the appliance, the name or the location without disconnecting the module. **Change the Wi-Fi network** is used after moving or changing the Wi-Fi password: it repeats the Bluetooth search with the same module (if the module cannot find its network, it offers Bluetooth by itself after 5 minutes). **Delete** unlinks the module, after confirmation; it stops sending data and must be linked again to be used.

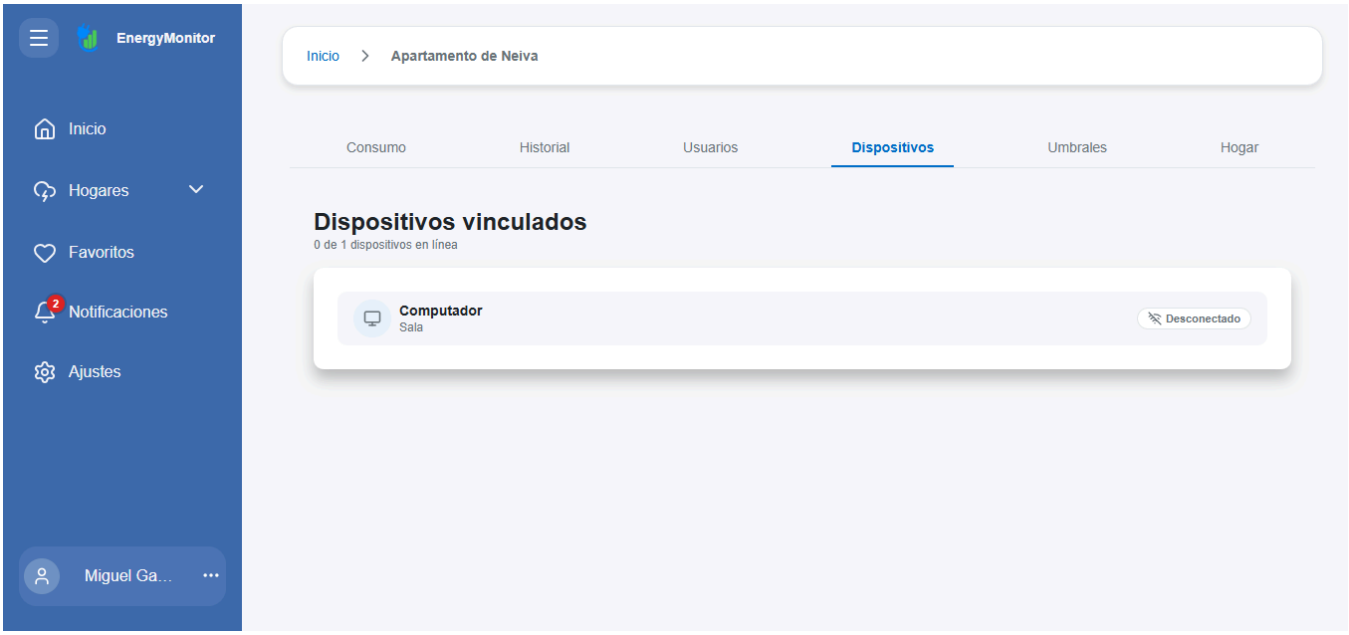
Expected result: The module appears Online and its first reading arrives in about a minute.

Notes:

- **Web:** Bluetooth linking works in Chrome or Edge on a computer and in Chrome on Android, with the web over HTTPS; Chrome on Linux needs "Experimental Web Platform features" turned on. Other browsers use the portal alternative: join the "EnergyMonitor-Setup" Wi-Fi network, open `http://192.168.4.1` and set the network there.
- **Mobile:** if Bluetooth is off or the permission is missing, the app says so; it is granted in the system settings.
- A module becomes Disconnected when it stops reporting for about 2.5 minutes; all members receive an alert.
- **Member:** sees the list without actions; if there are no modules, the screen says the owner has not linked devices yet.



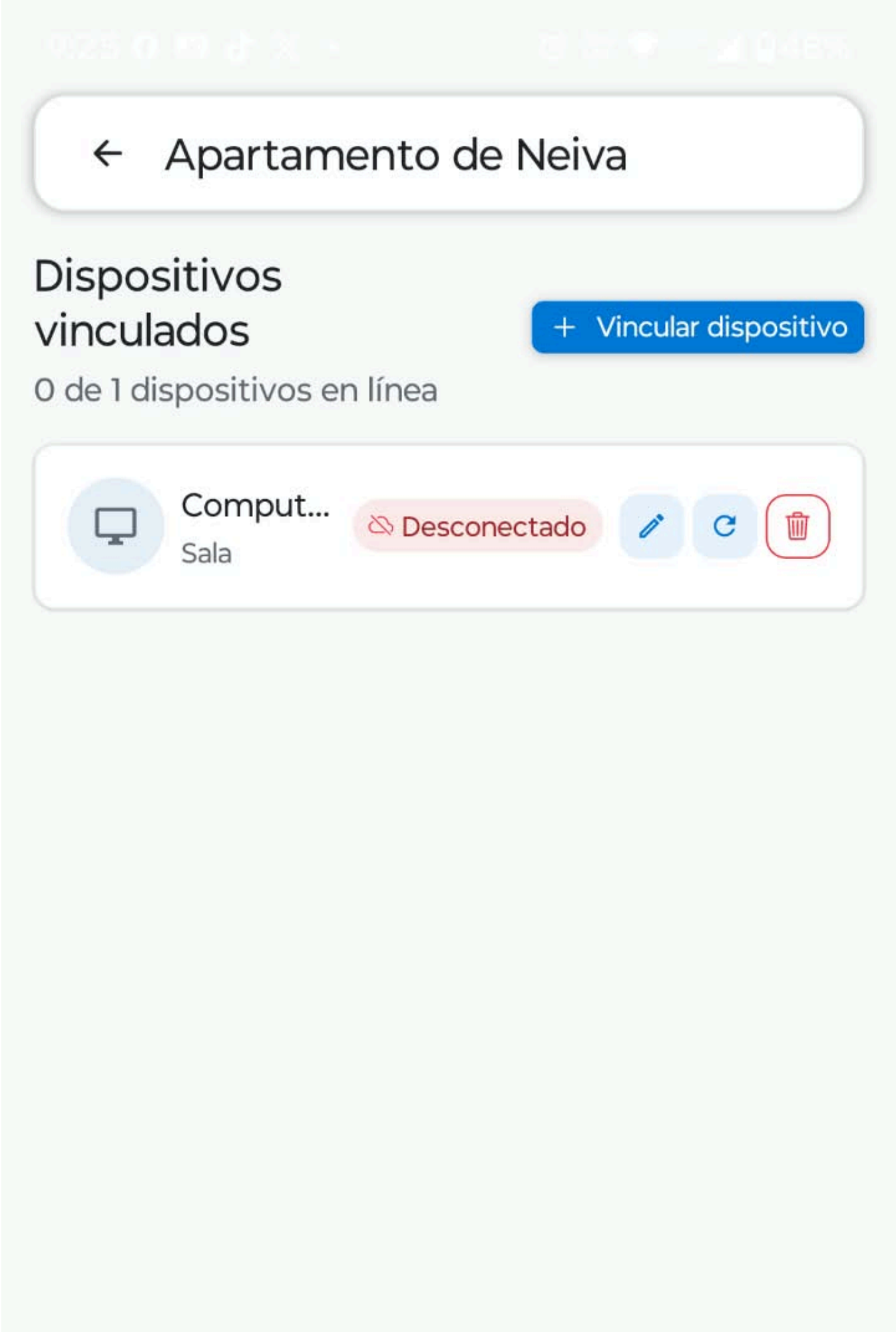
Devices, owner view (web)

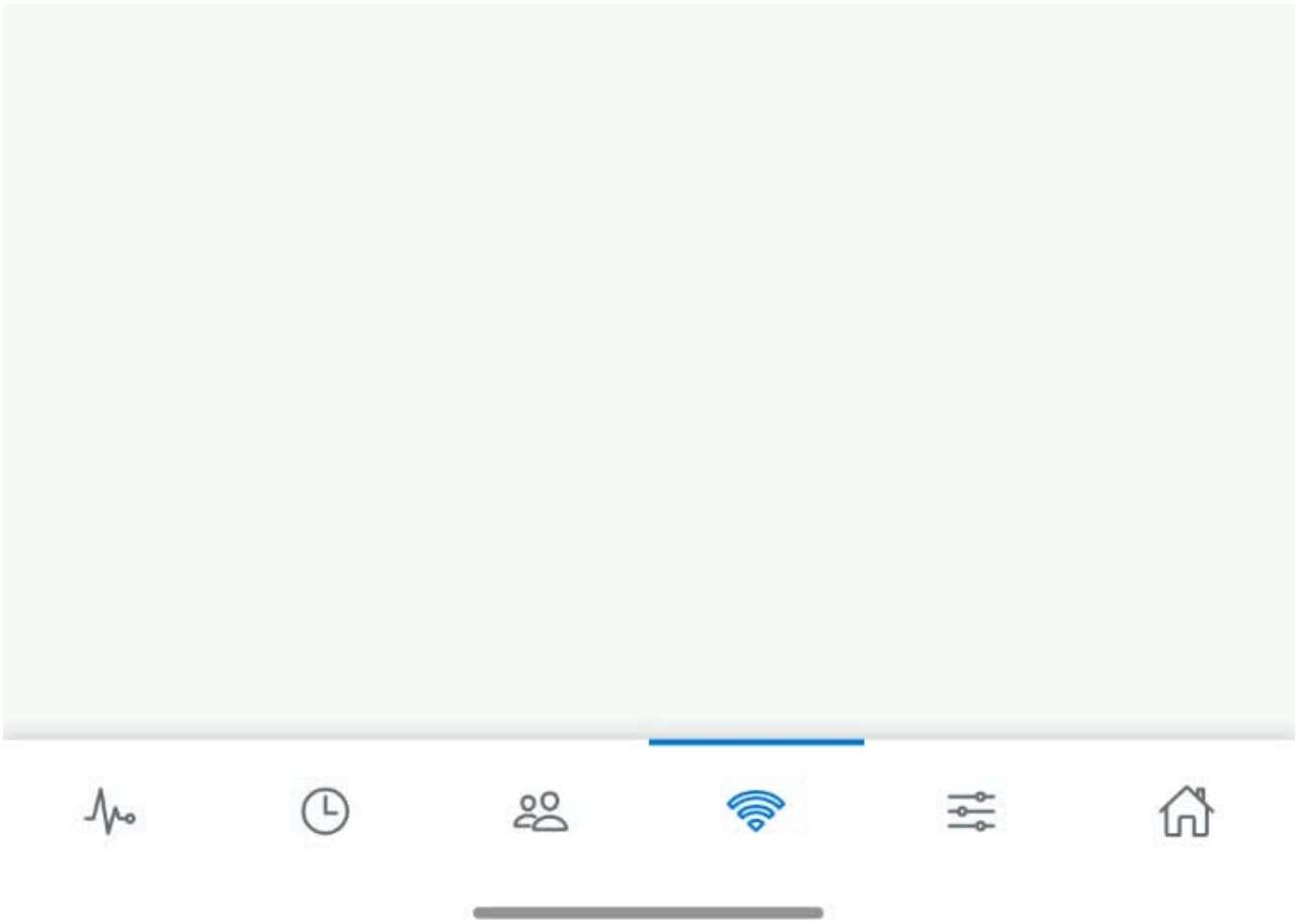


Devices, member view (web)



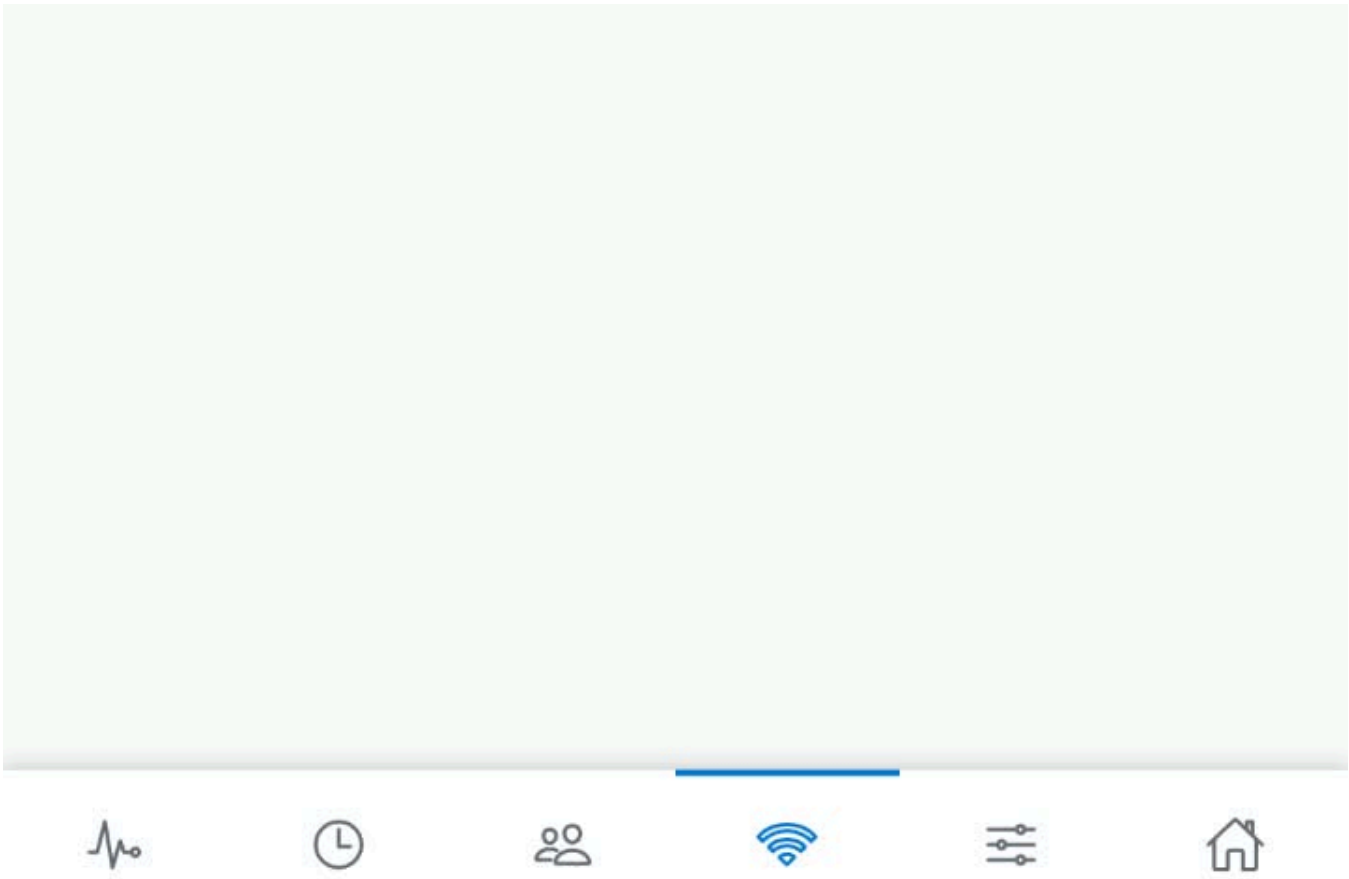
"Link device" wizard, Find your module step (web)


Apartamento de Neiva
Dispositivos vinculados
+ Vincular dispositivo
0 de 1 dispositivos en línea
Comput...
Sala
Desconectado



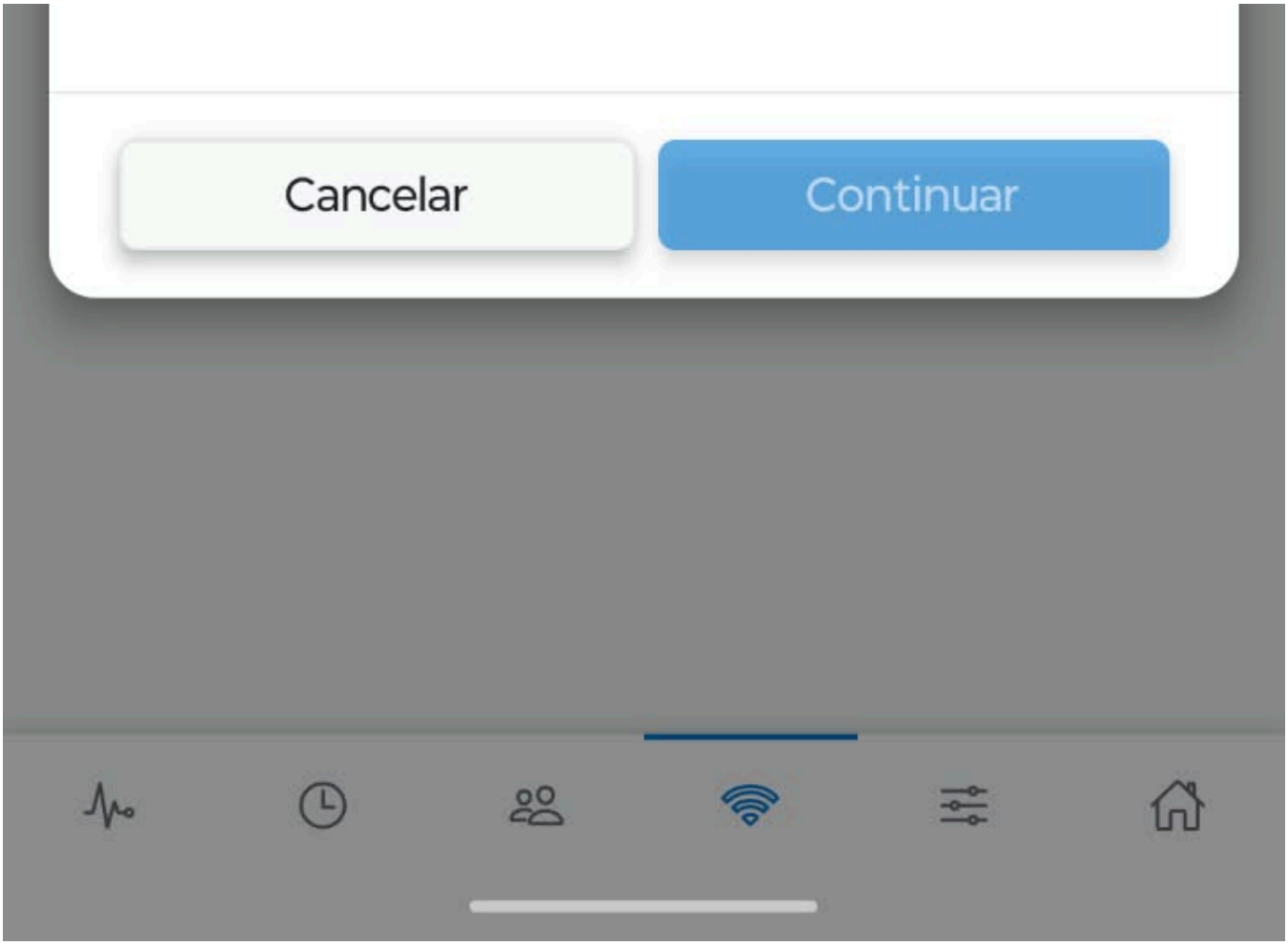
Devices, owner view (mobile)


Apartamento de Neiva
Dispositivos vinculados
0 de 1 dispositivos en línea
Computador
Sala
Desconectado



Devices, member view (mobile)


9:25
48%
Apartamento de Neiva
Dispositivos vinculados
+ Vincular dispositivo
0 de 1 dispositivos en línea
Vincular dispositivo
PASO 1 DE 5
Encuentra tu módulo
Enchufa el módulo y mantén pulsado su botón BOOT 3 segundos, hasta que la luz parpadee. Después pulsa «Buscar módulo» y elige el de la lista.
Acerca el módulo y pulsa buscar.
Buscar módulo



Bluetooth search and list of found modules (mobile)

9.5 Thresholds

Description: Sets the consumption limit of the home used for the limit alerts and some recommendations.

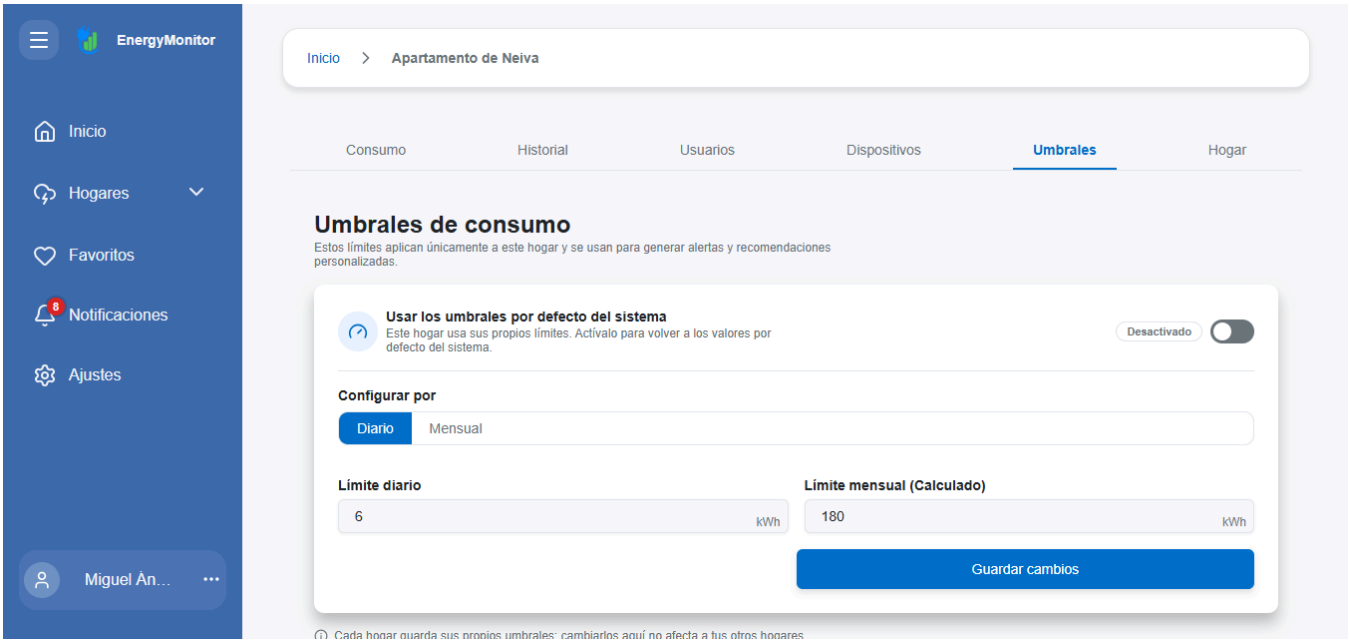
Field	Detail
Role that can access	Owner (edits) and members (view only).
Access path	Home detail > Thresholds.
Screen components	"Use the system default thresholds" switch (6 kWh per day and 180 kWh per month); "Configure by" Daily or Monthly selector; Daily limit and Monthly limit fields (the one not configured shows as "Calculated"); Save changes button.

Steps to use it (owner):

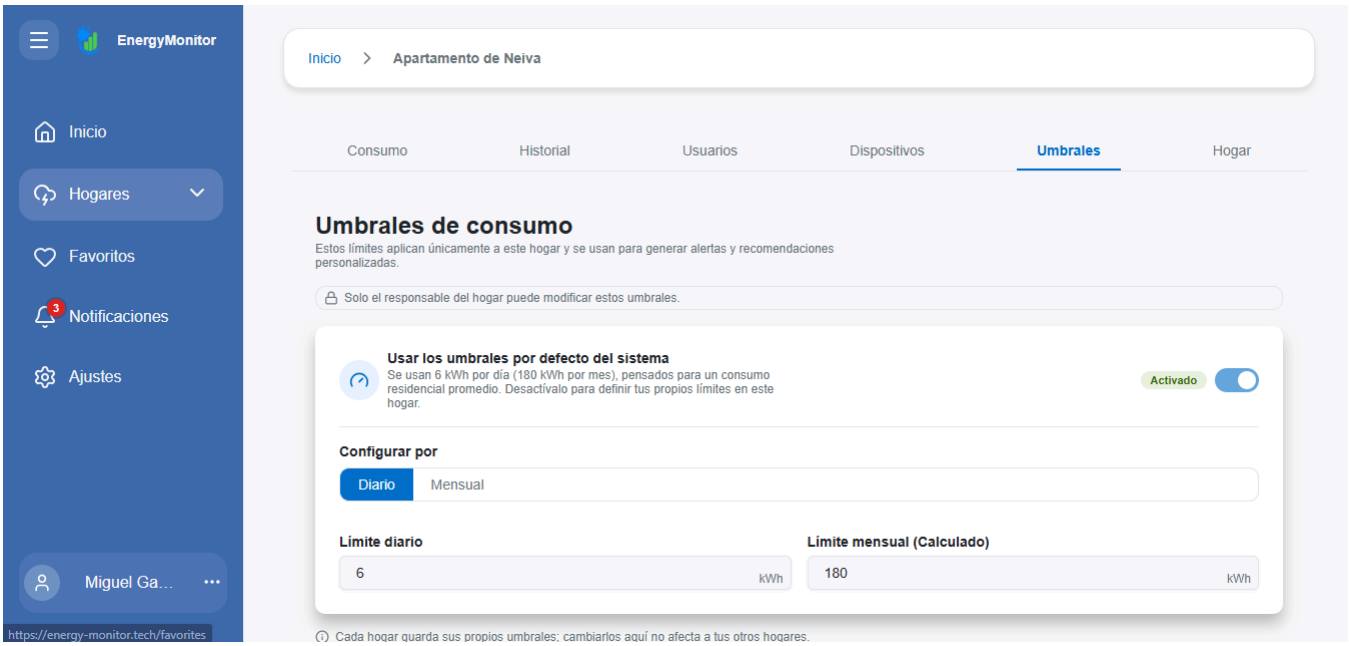
1. Open the Thresholds tab.
2. Turn off the switch to set your own limit.
3. Choose whether to configure it per day or per month and type the value in kWh (greater than 0). The other one is calculated, with 30 days per month.
4. Press Save changes. To go back to the system values, turn the switch on again.

Expected result: The new limit is saved and used for the "Daily/monthly limit exceeded" alerts. The other members receive an activity notification.

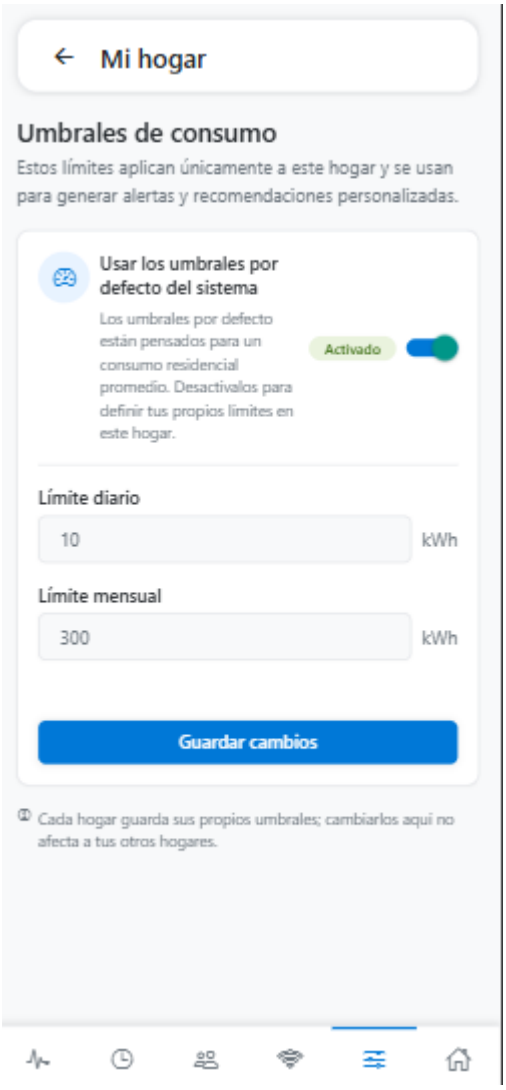
Notes: Each home has its own thresholds. The daily limit cannot exceed the monthly one. **Member:** sees the values and the notice "Only the owner of the home can change these thresholds".



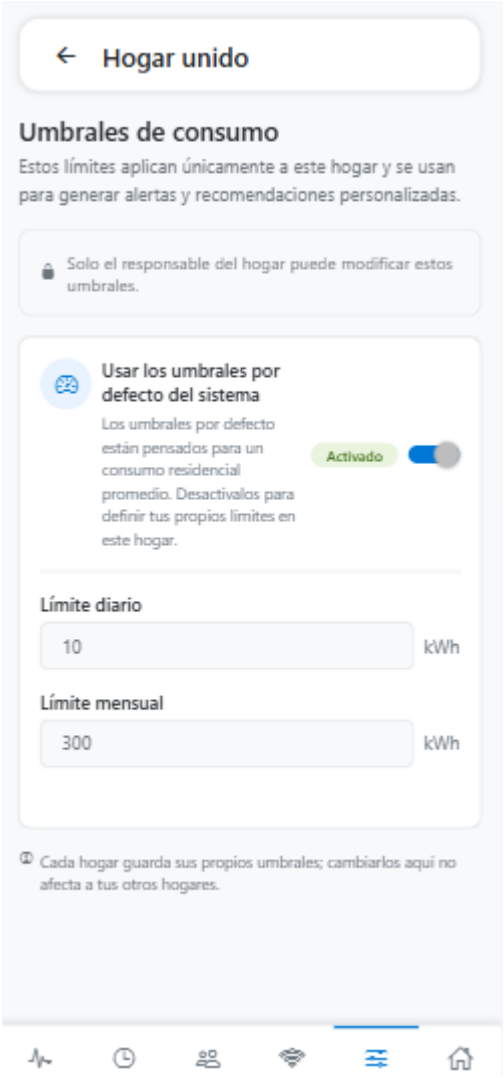
Thresholds, owner view (web)



Thresholds, member view (web)



Thresholds, owner view (mobile)



Thresholds, member view (mobile)

9.6 Home

Description: Gathers the information of the home, its owner and, for the owner, the access code.

Field	Detail
Role that can access	Owner and members.
Access path	Home detail > Home.
Screen components	Home information (name, type, address, description and creation date); owner with their email. **Owner:** access code with the Copy button and the Delete home button. **Member:** Leave home button.

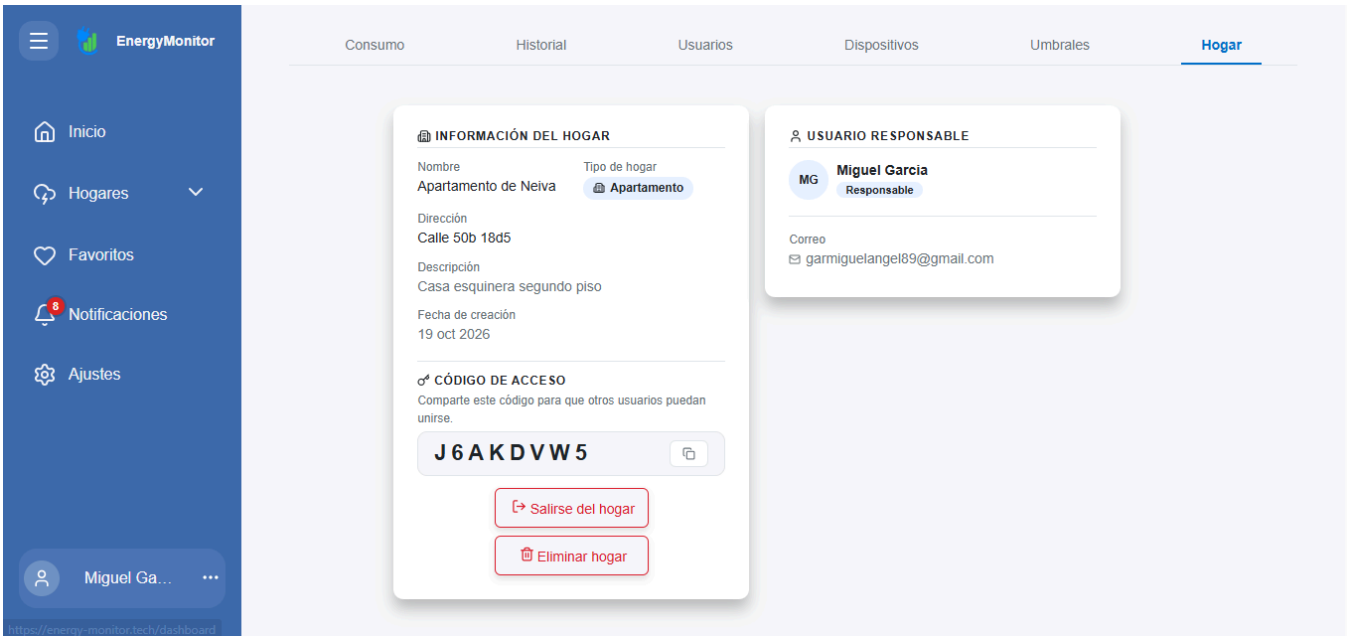
Steps to use it:

1. Open the Home tab and check the data.
2. **Owner – share:** press Copy access code and send it to whoever should join (section 8).
3. **Owner – delete:** press Delete home, type the name of the home to confirm and accept.
4. **Member – leave:** press Leave home and confirm.

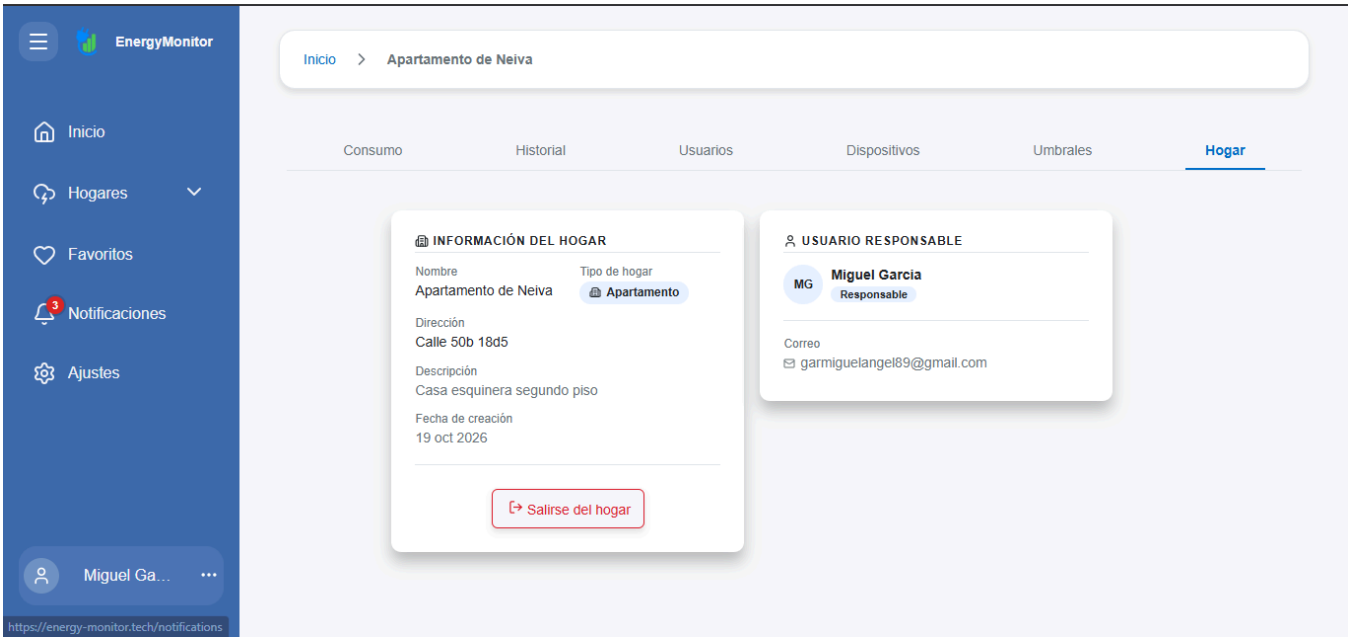
Expected result: The owner shares access or deletes the home; the member stops seeing the home after leaving.

Notes: Deleting a home is permanent: it disappears for all its members and its modules are unlinked, ready for another home. The owner cannot leave while being the only owner: first transfer the ownership (section 9.3).

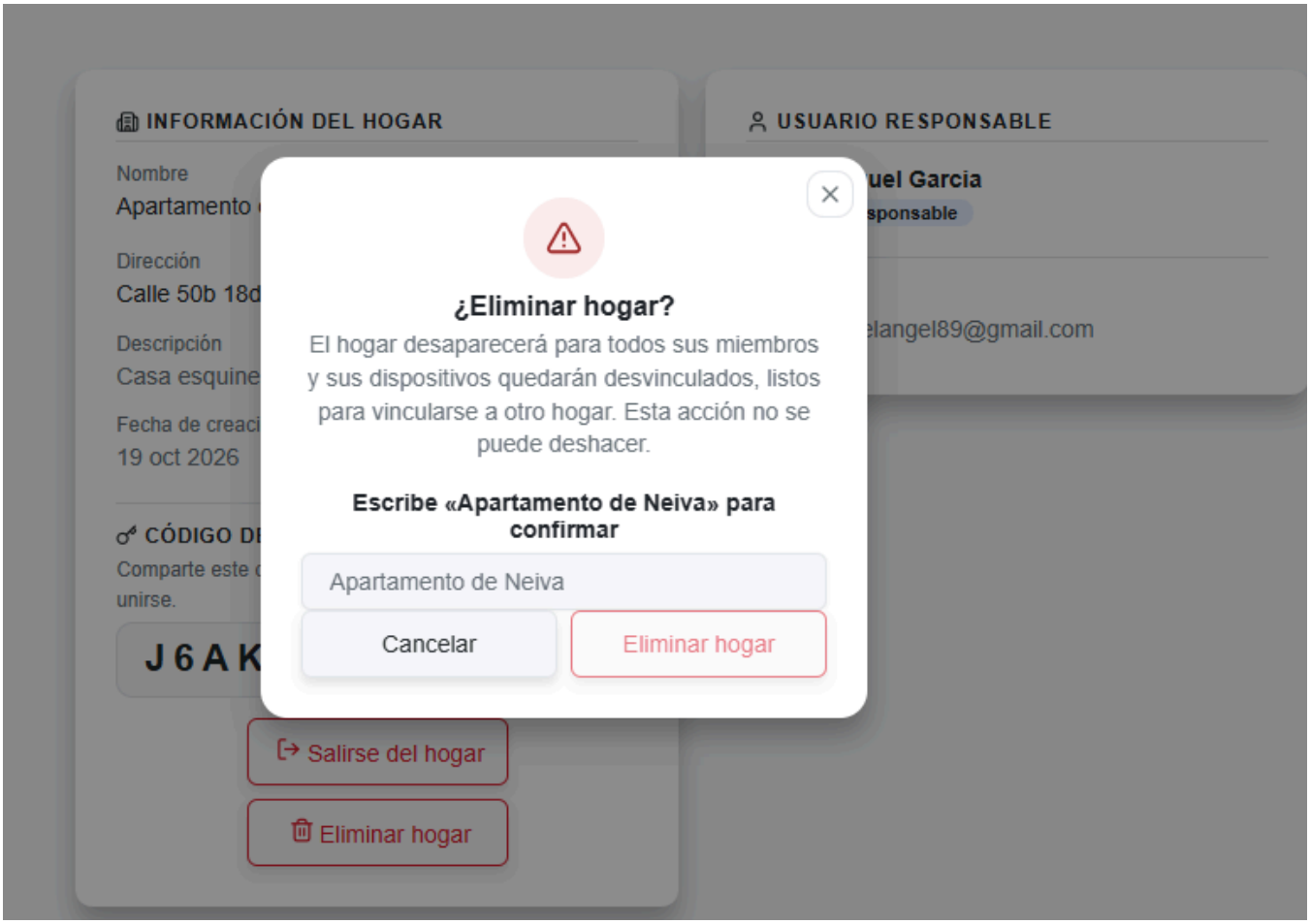
Whoever leaves can only come back with the access code.



Home, owner view (web)



Home, member view (web)



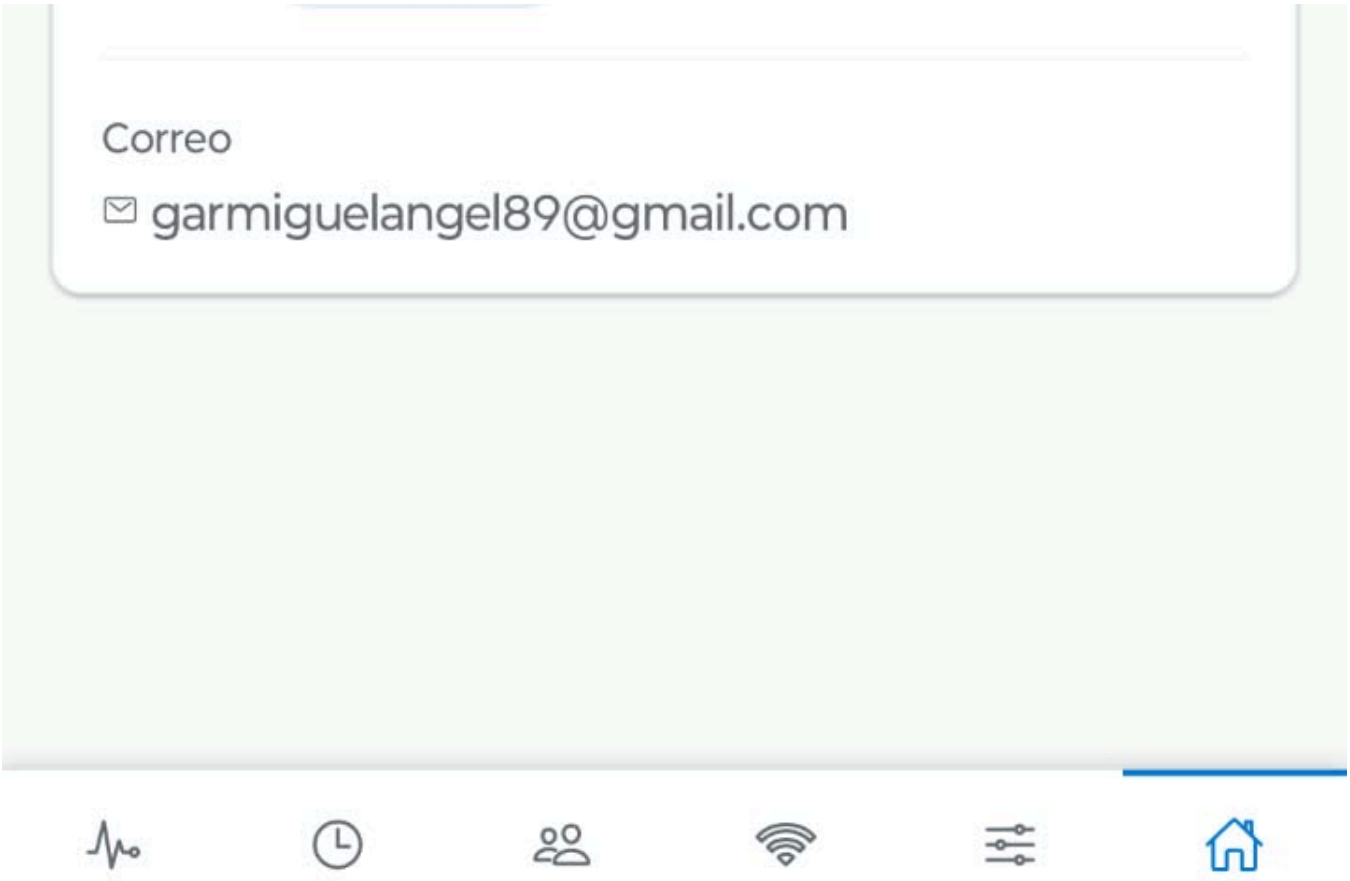
"Delete home" confirmation with the typed name (web)


Apartamento de Neiva
Información del hogar
Nombre
Apartamento de Neiva
Tipo de hogar
Apartamento
Dirección
Calle 50b 18d5
Descripción
Casa esquinera segundo piso
Fecha de creación
19 de oct de 2026
Código de acceso
Comparte este código para que otros usuarios puedan unirse.
J6AKDVW5
Salirse del hogar



Home, owner view (mobile)


Apartamento de Neiva
Información del hogar
Nombre
Apartamento de Neiva
Tipo de hogar
Apartamento
Dirección
Calle 50b 18d5
Descripción
Casa esquinera segundo piso
Fecha de creación
19 de oct de 2026
Salirse del hogar
Usuario responsable
M
Miguel Garcia
Responsable



Home, member view (mobile)

10. Favorites

[Web and mobile]

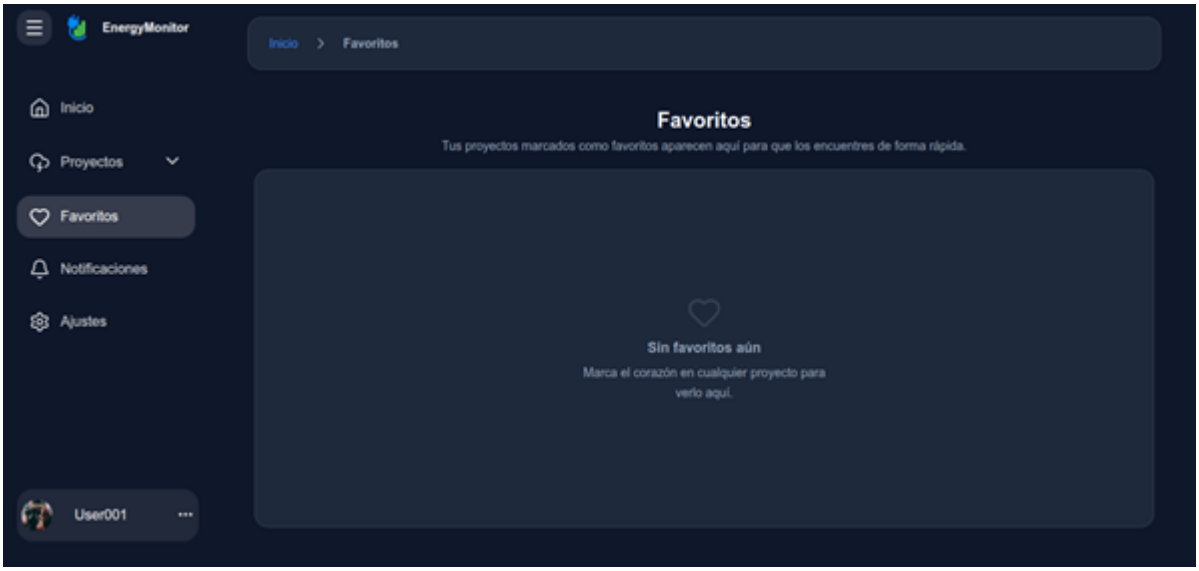
Description: Quick access to the homes you check most often.

Field	Detail
Role that can access	User (not administrator).
Access path	Side menu > Favorites.
Screen components	Cards of the homes marked with the heart, or an empty state when there are none; loading indicator and error message with a retry option.

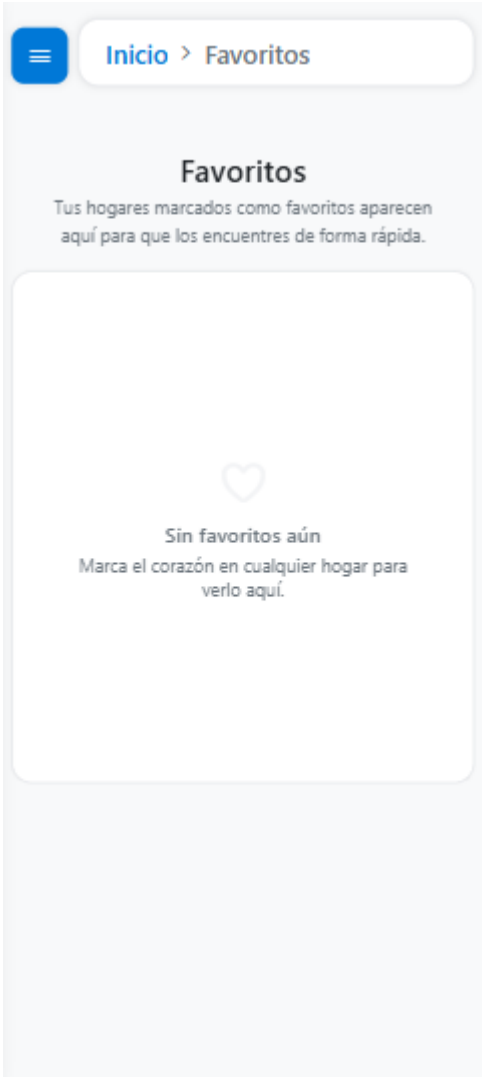
Steps to use it:

1. On the dashboard, press the heart of a home card.
2. Open Favorites from the side menu.

Expected result: The user quickly finds their favorite homes.



Favorites (web)



Favorites (mobile)

11. Notifications

[Web and mobile]

Description: Gathers the alerts, the saving recommendations and the activity of the account and of its homes.

Field	Detail
Role that can access	User. (On the web, for the administrator this option shows the administration notices: section 15.5; the mobile app has no administration notices: an administrator sees notifications like any other user.)
Access path	Side menu > Notifications (shows the number pending).
Screen components	**All**, **Alerts**, **Recommendations** and **Activity** tabs, each with its counter; filter by home (All, My account or one home); Mark all as read; button to delete the resolved or read ones; cards with icon, title, message, home, device, date, status (New, Active, Resolved or Read) and actions.

What each tab receives:

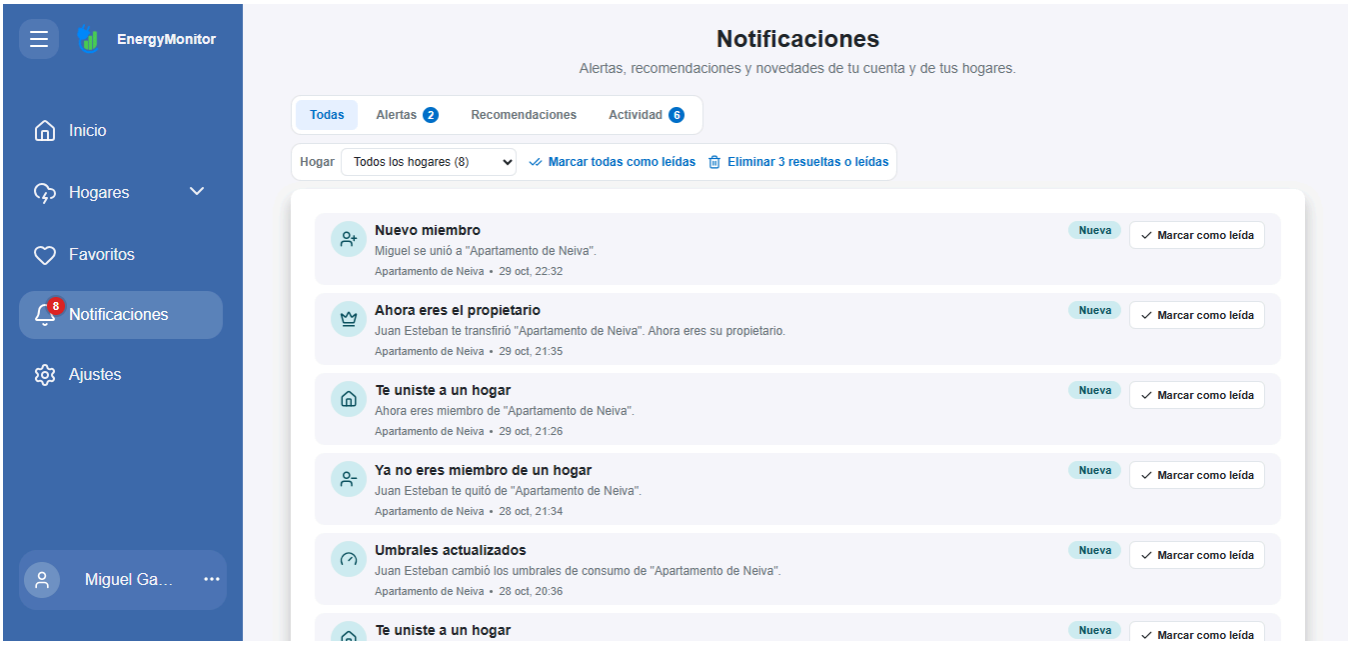
Tab	Content
Alerts	High or critical consumption of a device; daily or monthly limit reached; device disconnected; new device linked and sending measurements. Consumption and connection alerts resolve themselves when the situation returns to normal.
Recommendations	Advice generated every hour: shift consumption away from the peak hour (18:00–22:00), turn off a device that draws power at night, a week well above the previous one, a pace that will exceed the home's limit and a device that consumes more than before. The same advice is not repeated within 7 days.
Activity	Welcome to the account; home created; someone joined, left or was removed; thresholds changed; ownership transferred; home deleted; device linked or unlinked. They are read in the chosen language, even if the home no longer exists.

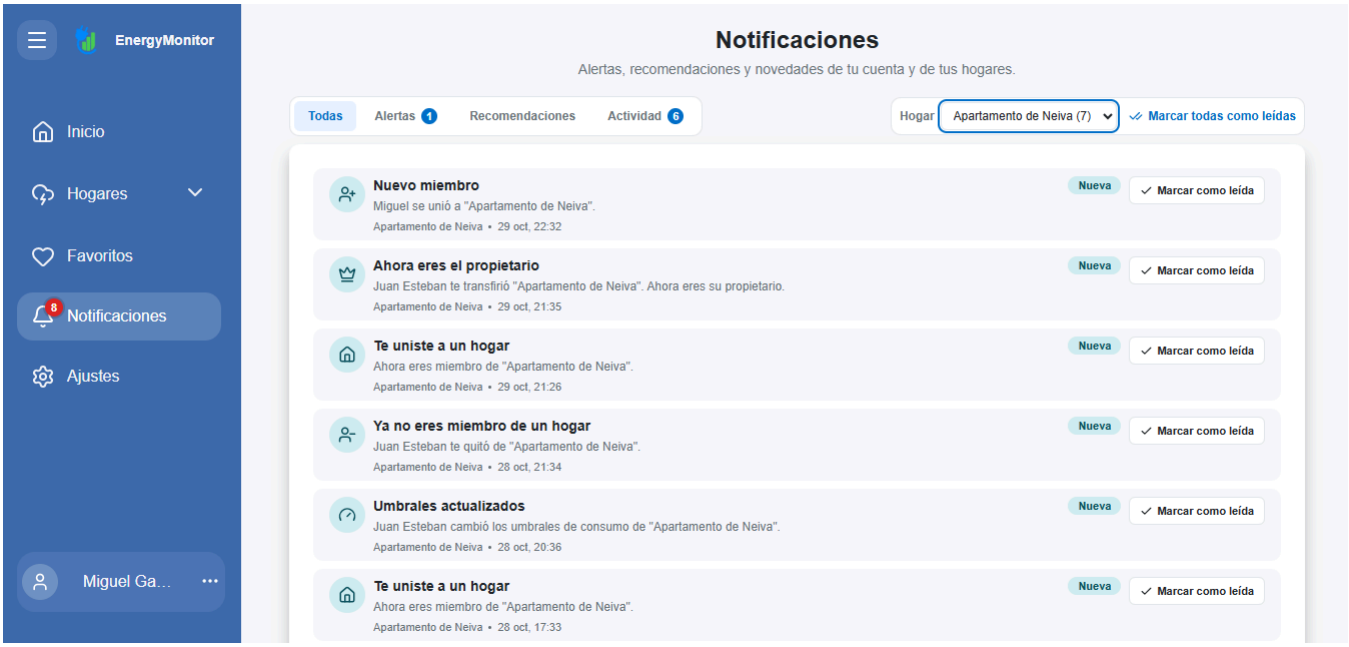
Steps to use it:

> 1. Open Notifications.
> 2. Choose a tab and, if you wish, filter by home.
> 3. Mark one or all as read; delete the ones no longer needed (alerts once resolved; recommendations once read).

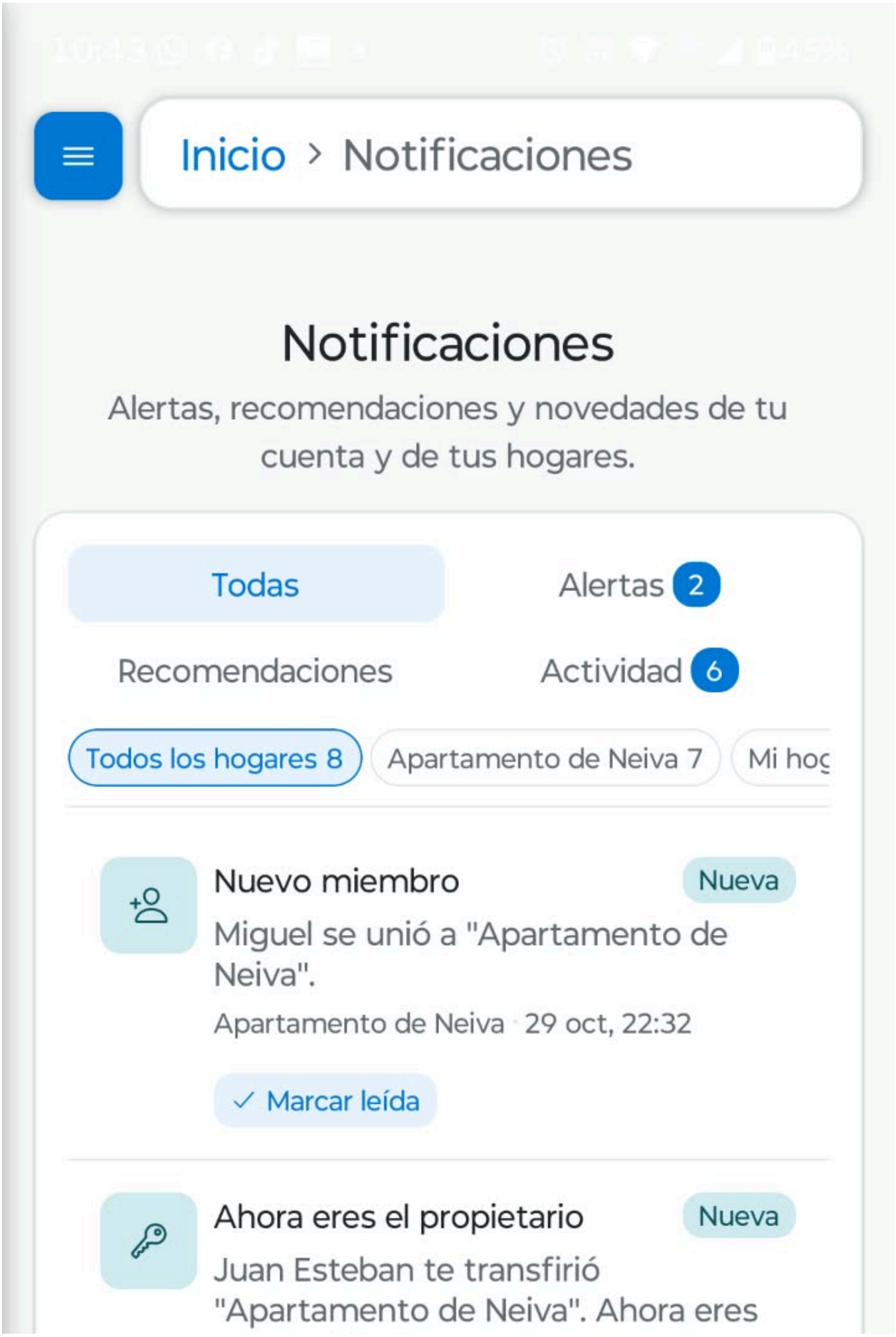
Expected result: The user learns in time what happens in their homes and acts on it.

Notes: Alerts and recommendations can also arrive by email and as push notifications, according to the settings (section 12). Activity always appears in the inbox. **Unusual consumption:** if an appliance consumes far more than usual (at least 200 W and 2.5 times its hourly average of the last 7 days, with at least 24 hours of history), the home receives a threshold alert that resolves by itself when consumption returns to normal. **Mobile:** the alert arrives, but with the generic threshold text; the specific text is pending.



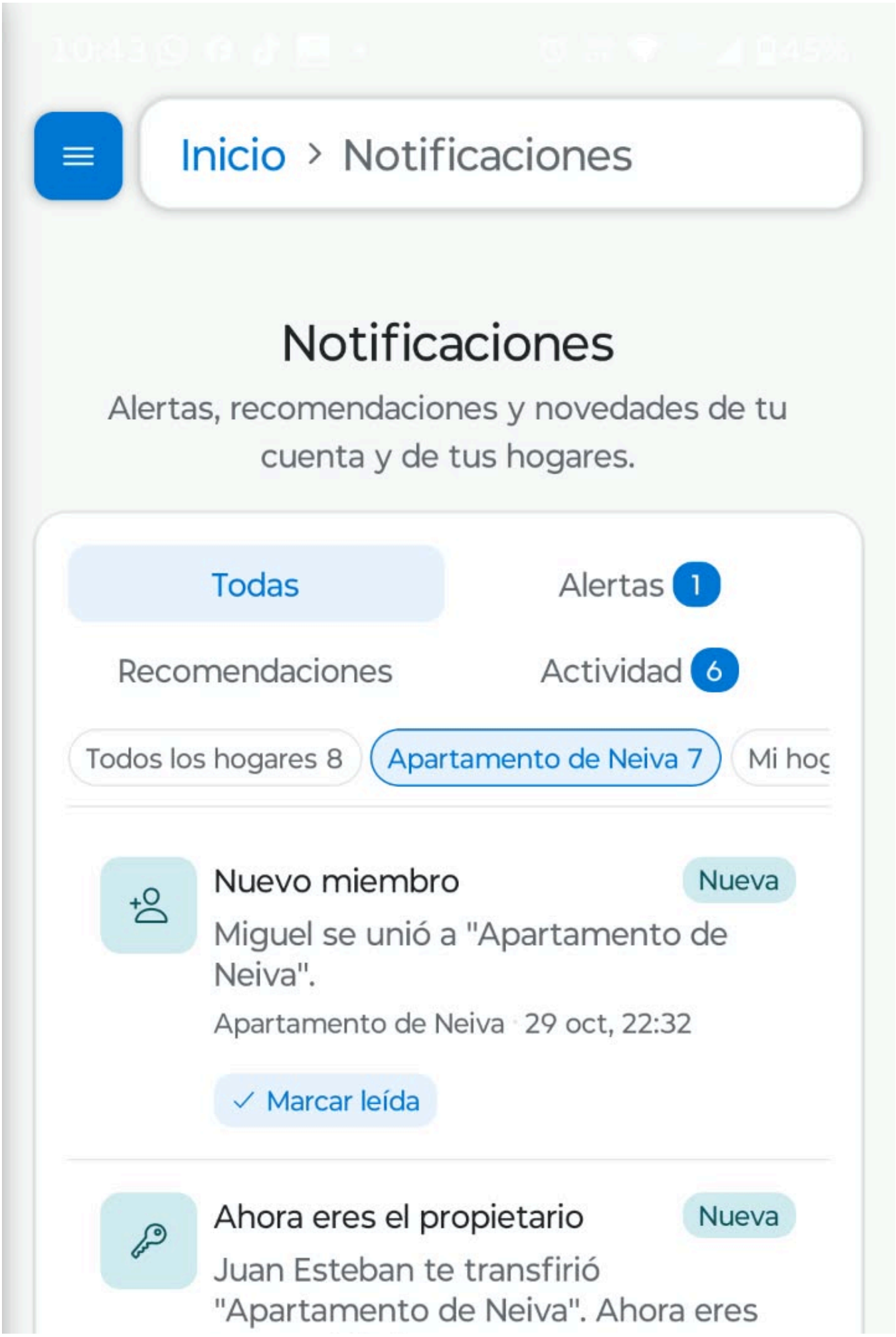


Activity tab with home filter (web)


Inicio › Notificaciones
Notificaciones
Alertas, recomendaciones y novedades de tu cuenta y de tus hogares.
Todas
Alertas 2
Recomendaciones
Actividad 6
Todos los hogares 8
Apartamento de Neiva 7
Mi hoç
Nuevo miembro
Nueva
Miguel se unió a "Apartamento de Neiva".
Apartamento de Neiva · 29 oct, 22:32
✓ Marcar leída
Ahora eres el propietario
Nueva
Juan Esteban te transfirió "Apartamento de Neiva". Ahora eres



Notifications (mobile)


Inicio › Notificaciones
Notificaciones
Alertas, recomendaciones y novedades de tu cuenta y de tus hogares.
Todas
Alertas 1
Recomendaciones
Actividad 6
Todos los hogares 8
Apartamento de Neiva 7
Mi hoç
Nuevo miembro
Nueva
Miguel se unió a "Apartamento de Neiva".
Apartamento de Neiva · 29 oct, 22:32
✓ Marcar leída
Ahora eres el propietario
Nueva
Juan Esteban te transfirió "Apartamento de Neiva". Ahora eres



Activity tab with home filter (mobile)

12. Settings

[Web and mobile]

Description: Language, theme and notification preferences, and help center.

Field	Detail
Role that can access	User and administrator (on the web the administrator does not see the notifications section; in the mobile app the setting is shown as for a regular user, because the app does not distinguish the role).
Access path	Side menu > Settings. **Mobile:** the screen has the title "Settings".
Screen components	**System language:** Español, English, Português or Français. **System themes:** Light · Default, Light · Theme 2, Dark · Theme 1 and Dark · Theme 2. **Notifications:** Email, Push notifications and Combined mode (both at once). **Help center:** Documentation, Technical support and What's new (marked "Coming soon").

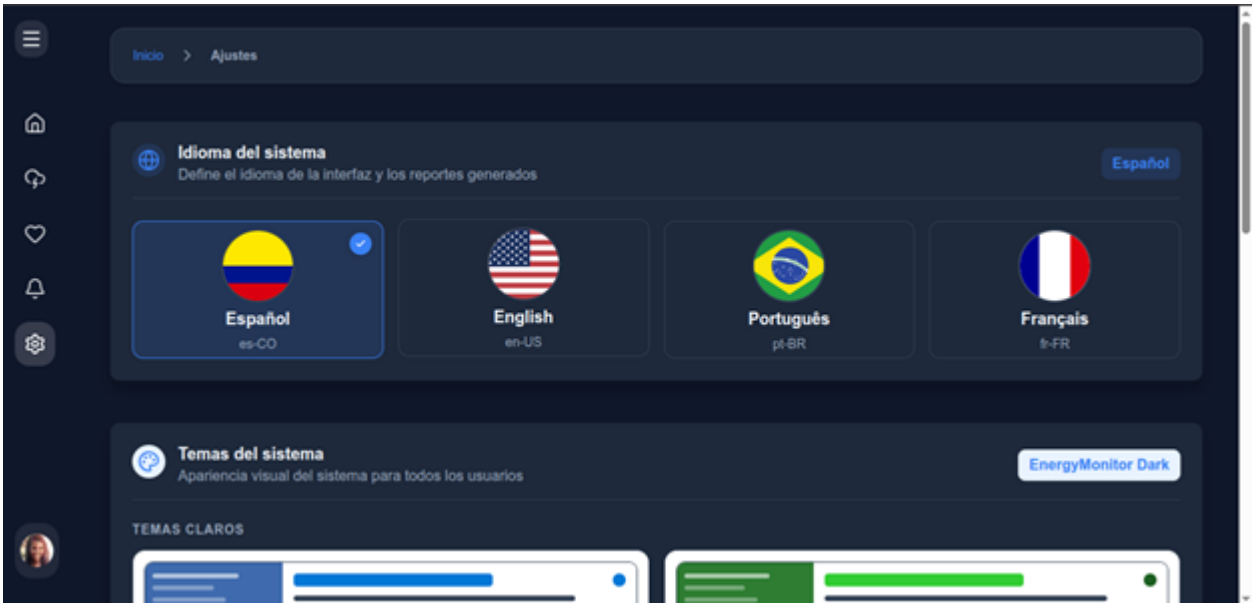
Steps to use it:

1. Open Settings.
2. Choose language and theme: the change applies at once.
3. Turn email and push on or off.

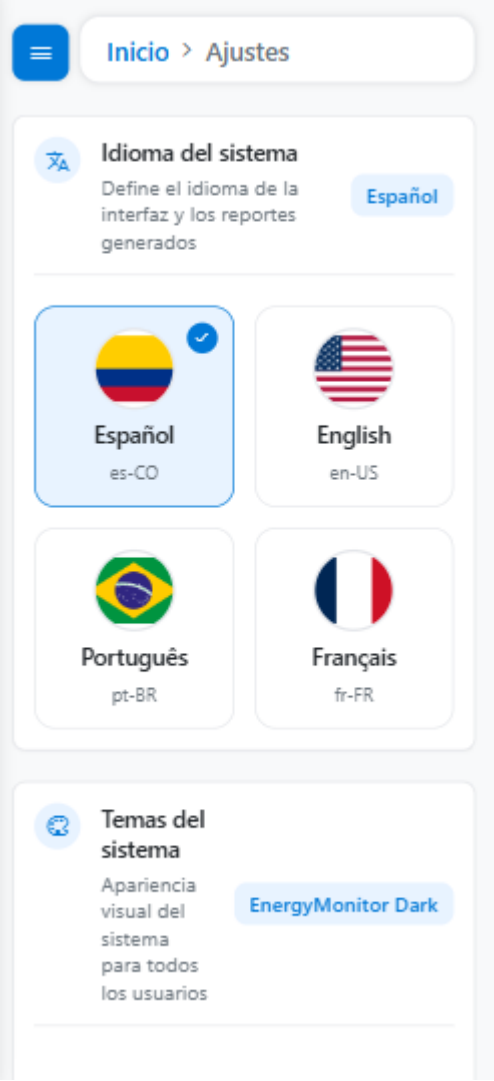
Expected result: The preferences are reflected across the whole application.

Notes:

- **Web:** the theme is stored in that browser; push reaches that browser even with the tab closed. If the browser blocked notifications, turn them on from the padlock of the address bar. On iPhone, install the web on the home screen.
- **Mobile:** push reaches the phone and the subscribed browsers of the same account; if Android lacks the permission, the app says so.
- If the server does not have push configured, the option appears as unavailable.
- **Brave (web):** this browser ships with the Google service that delivers push turned off. If turning it on does not work, the screen says to go to brave://settings/privacy, turn on «Use Google services for push messaging», restart Brave and turn push on again.



Settings (web)



Settings (mobile)

12.1 Help center

[Web only] On mobile the cards are shown but they do not open any window yet.

Below the preferences, Settings has three cards. Each opens a window that is closed with **Close**:

Card	What it shows	Who sees it
Documentation	Usage guides and feature reference according to the role of the account: user or administrator.	User and administrator
Technical support	The available administrators. You choose whom to write to and **Create ticket** opens the email with the subject already written; the team answers by email.	Only the user: the administrator handles support and does not see the card
News	Improvements, new features and fixes grouped by date, with the version and the commit that is online. They are generated automatically on every deployment.	User and administrator

Note: the news titles come in the language of the commits (English) even if the interface is in another language.

13. My account

[Web and mobile]

Description: Personal data of the account, photo, password and account deletion.

Field	Detail
Role that can access	User and administrator.
Access path	**Web:** profile (bottom of the side menu) > My account (`/account`). **Mobile:** side menu > My account; the screen has the title "My account".
Screen components	Profile photo with the Add, View and Delete photo options; First name, Last name and Email fields, each with its Edit button; Password with Change password; Do you want to delete your account? with Delete account.

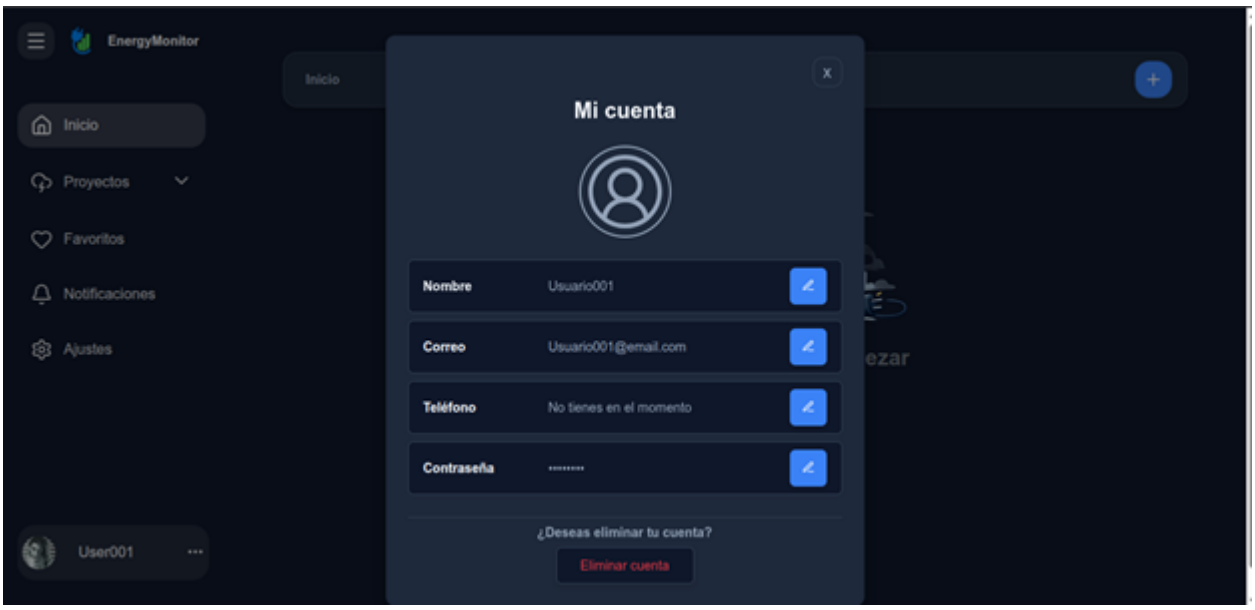
Steps to use it:

1. Open My account.
2. Press Edit on the data to change, type the new value and Save. The last name is optional.
3. **Change password:** type the current one, the new one and its confirmation.
4. **Delete account:** press Delete account and confirm with the password. On a Google account, press **Send me the code**, type the 6-digit code received (it expires in 15 minutes) and confirm.

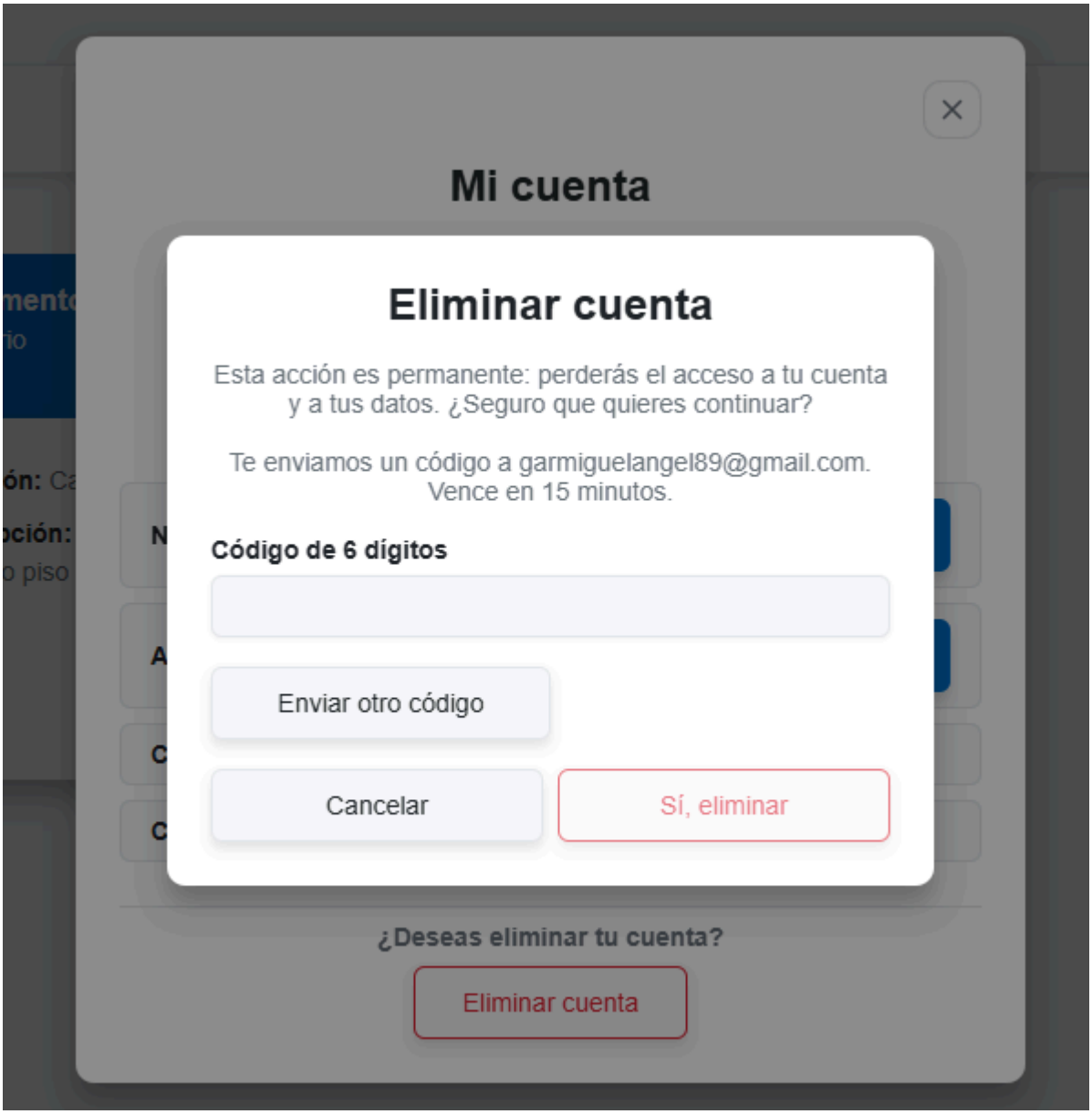
Expected result: The data is updated ("Profile updated."), or the account is deleted and the session closes ("Your account was deleted.").

Notes:

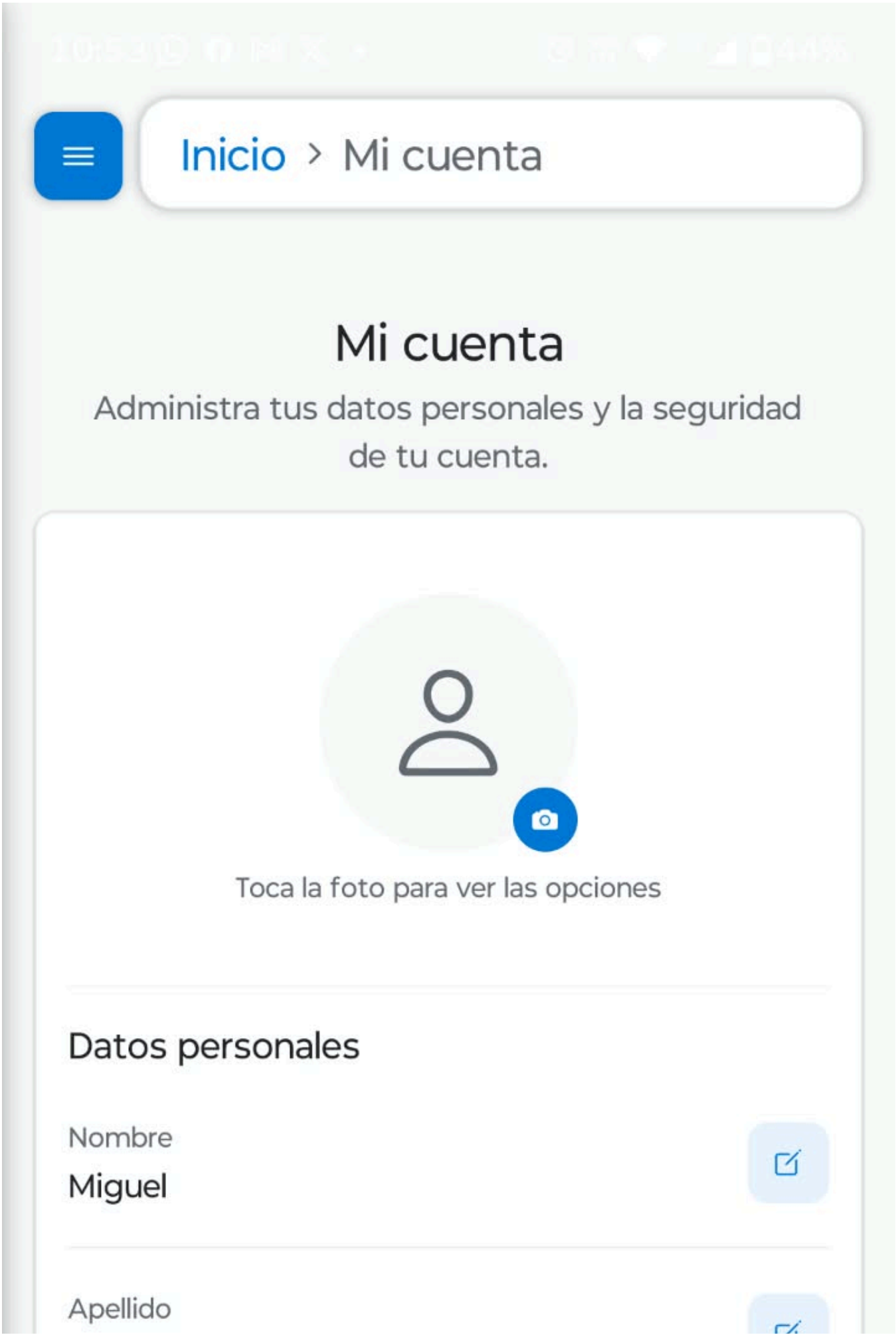
- Google accounts have no password of their own: the screen shows "You sign in with Google: your password is managed by Google" and does not offer the password change.
- If the email changes, the new address must be verified with the code it receives before signing in again.
- The account cannot be deleted while it is the only owner of a home with other members: first transfer the ownership or remove the members. Homes without other members are deleted with the account.
- **Mobile:** to choose a photo, the app asks for access to the gallery.
- With **View photo** the image opens in a viewer inside the application (on the web, inside the page without leaving it).

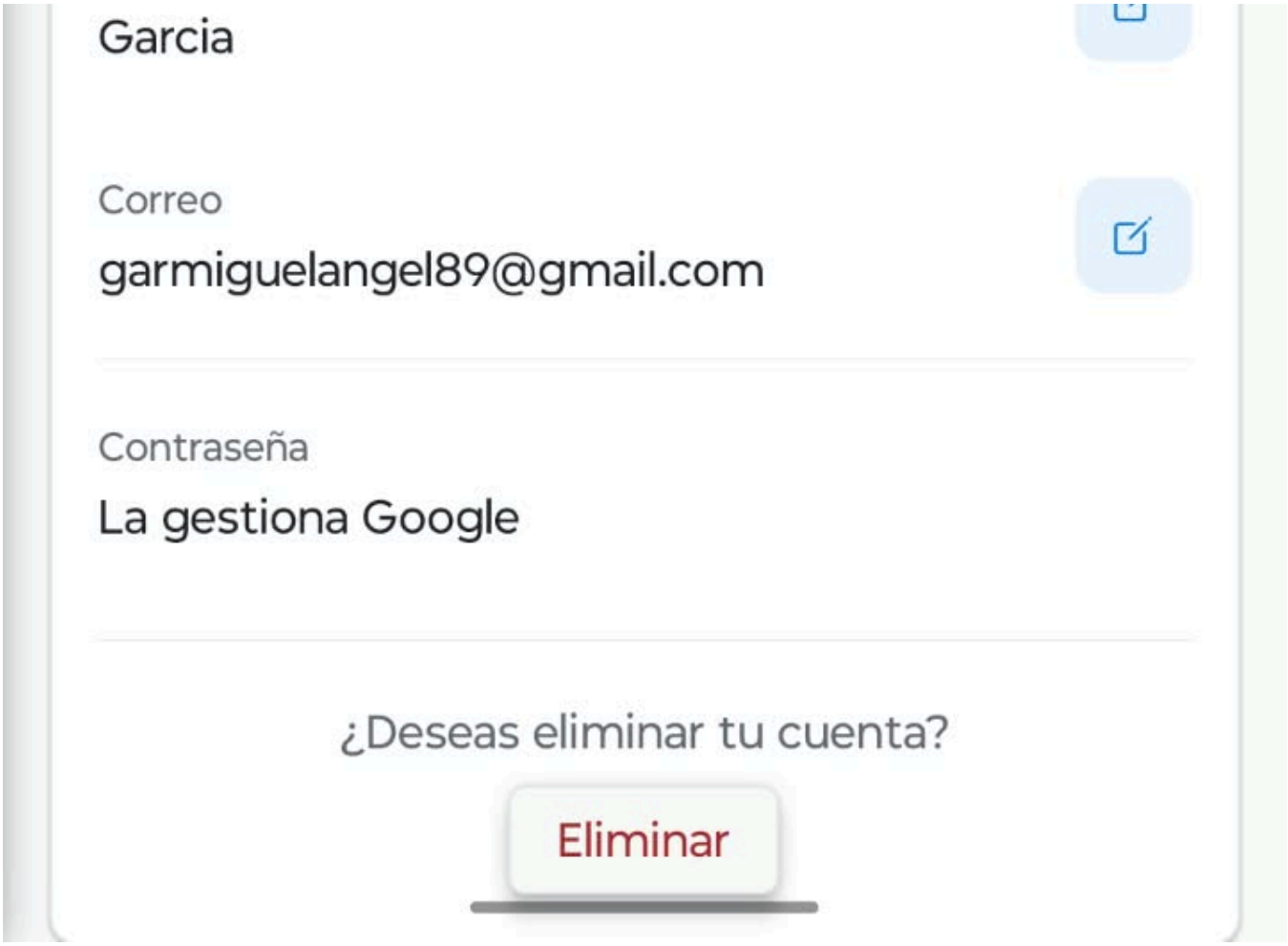


My account (web)



Deleting a Google account with a code (web)


Inicio > Mi cuenta
Mi cuenta
Administra tus datos personales y la seguridad de tu cuenta.
Toca la foto para ver las opciones
Datos personales
Nombre
Miguel
Apellido



My account (mobile)

14. Sign out

[Web and mobile]

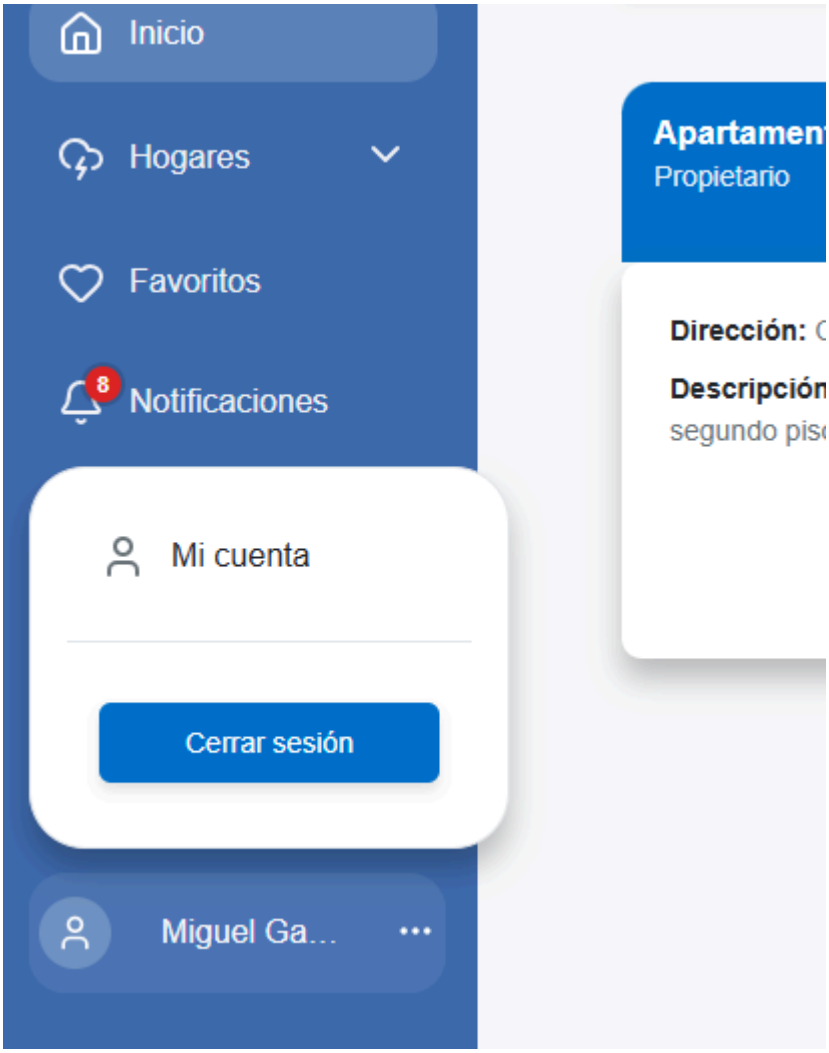
Description: Leaves the system safely.

Field	Detail
Role that can access	User and administrator.
Access path	**Web:** profile at the bottom of the side menu > Sign out. **Mobile:** side menu > Sign out.
Screen components	Sign out option.

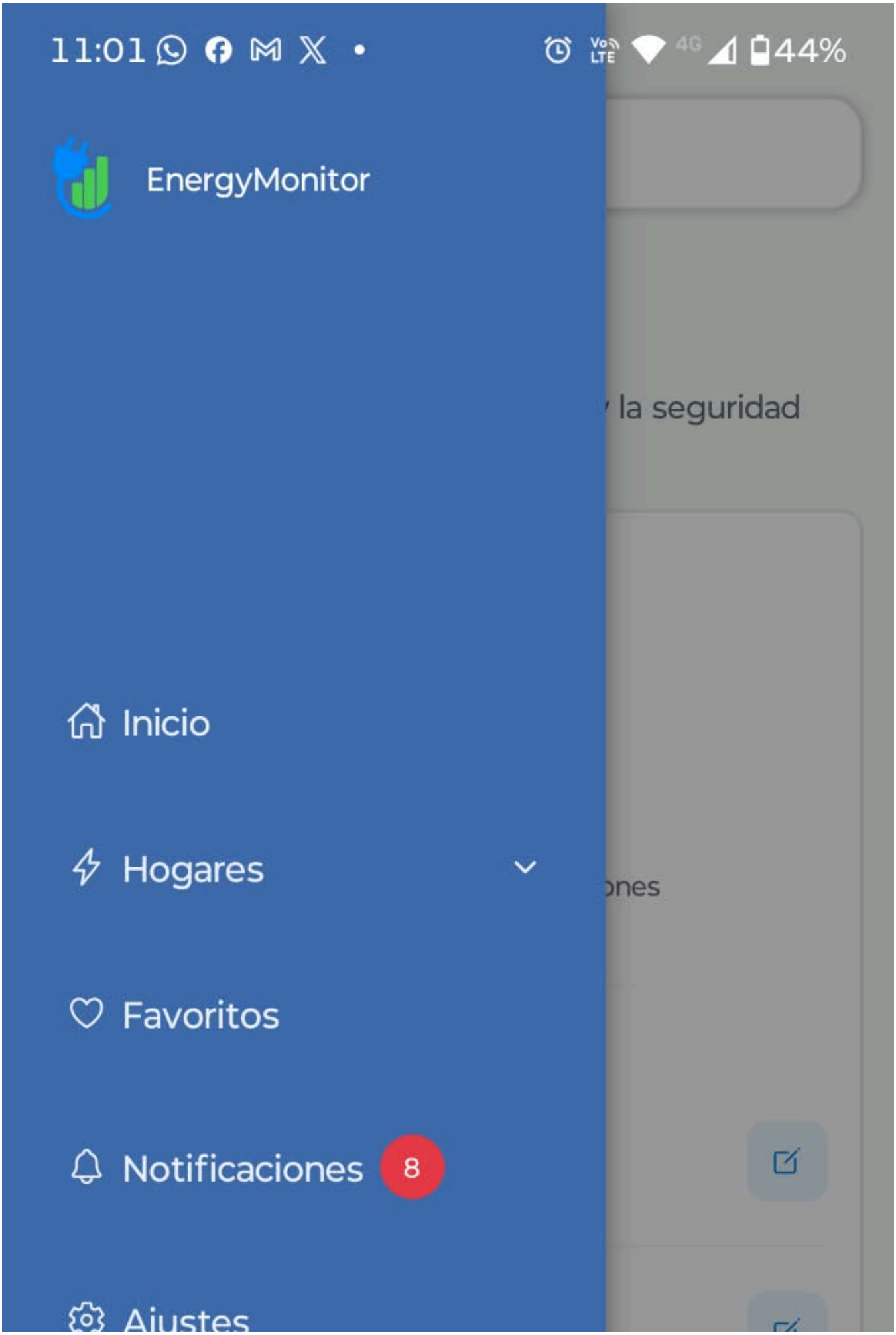
Steps to use it:

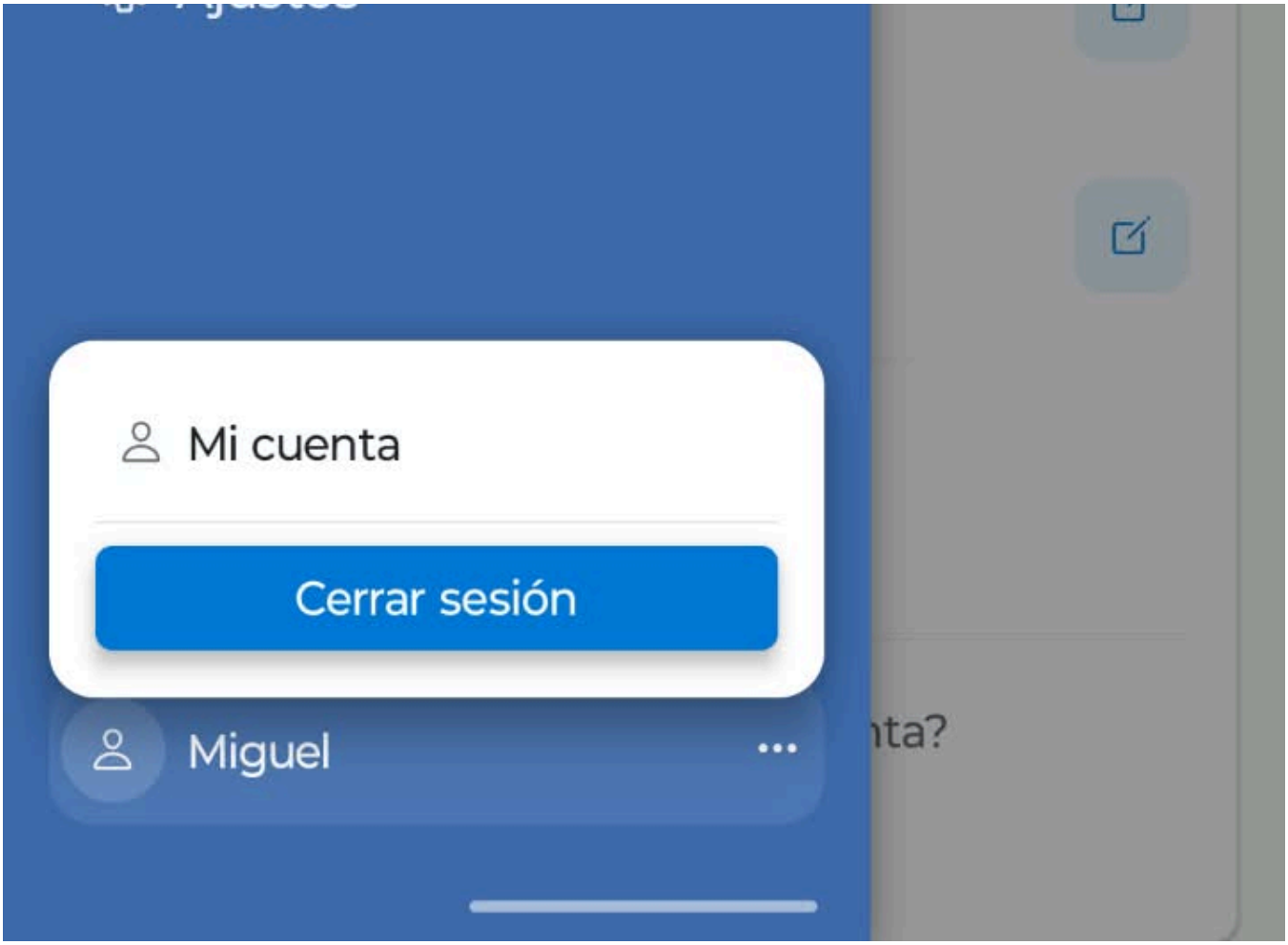
1. Open the profile menu (web) or the side menu (mobile).
2. Press Sign out.

Expected result: The session closes and the application returns to the sign-in screen (or to the landing page on the web).


Inicio
Hogares
Favoritos
8
Notificaciones
Mi cuenta
Cerrar sesión
Miguel Ga...
Apartamen
Propietario
Dirección:
Descripción
segundo pis

Sign out (web)


11:01
44%
EnergyMonitor
la seguridad
Inicio
Hogares
Favoritos
Notificaciones
8
Ajustes



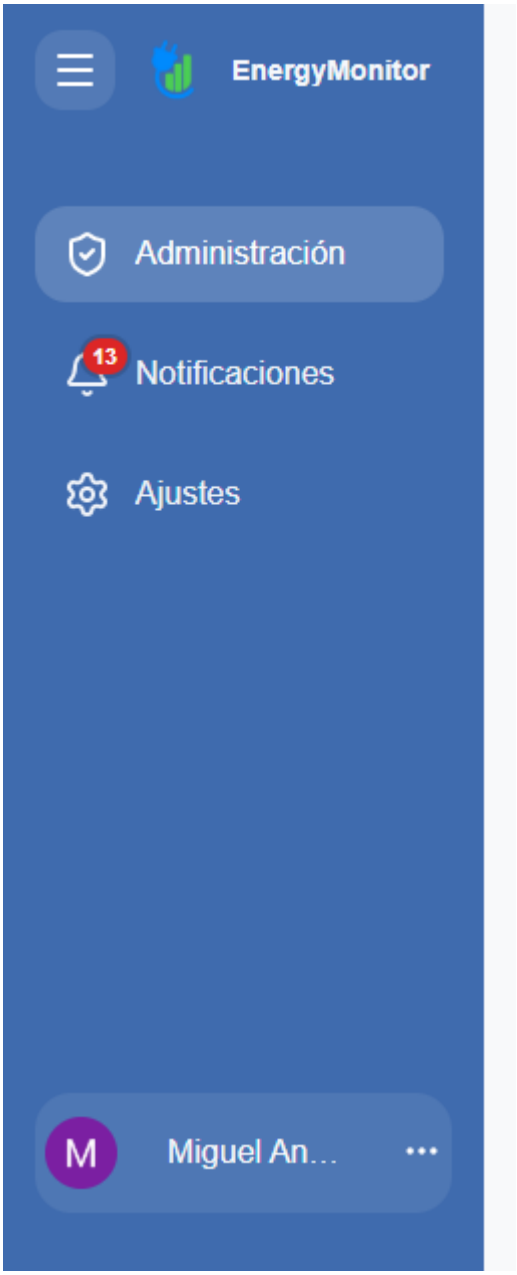
Sign out (mobile)

15. Administration (web only, administrator role)

[Web only] The mobile app has no administration screens: an administrator manages the platform from the web. The technical team grants the role the first time (administrator manual) and afterwards other administrators do it from this screen.

15.1 Access and administrator menu

An account with the administrator role that signs in on the web goes straight to **Administration**. Its side menu shows only **Administration**, **Notifications** and **Settings**, plus My account and Sign out. The home screens (dashboard, home detail and favorites) are not available to this account: if it tries to open them, the application sends it back to the administration. An account without permission that tries to open the administration goes back to its homes dashboard.



Side menu of an administrator account (web)

15.2 Platform users

Description: Directory of the platform accounts.

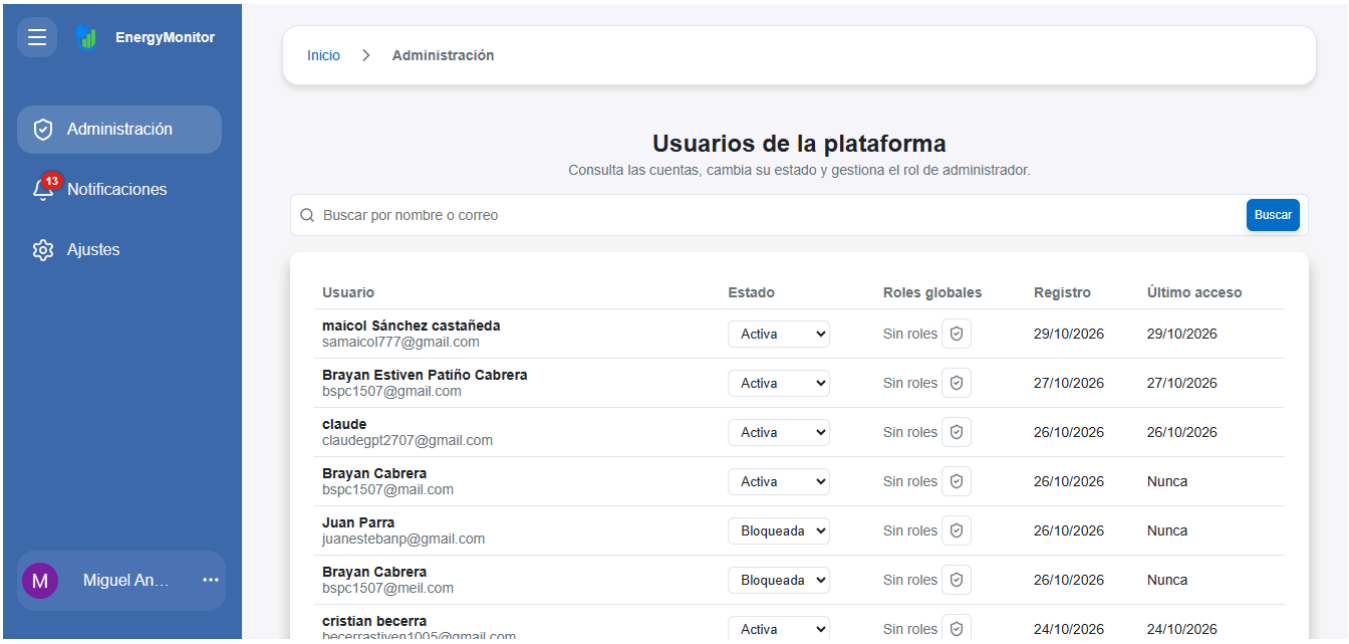
Field	Detail
Role that can access	Administrator.
Access path	Side menu > Administration (`/admin/users`).
Screen components	"Platform users" title; search by name or email with the Search button; table with the columns User (name, email and the "You" mark), Status, Global roles (or "No roles") with the shield button, Registered and Last access (or "Never"); pagination of 20 per page.

Steps to use it:

> 1. Open Administration.
> 2. To search, type part of the name or the email and press Search; clearing the search box brings the full list back.

> 3. With more than 20 results, use Previous page and Next page.

Expected result: The administrator finds any account and sees its status and roles.



Platform users (web)

15.3 Change the status of an account

Description: Activates, deactivates or blocks an account.

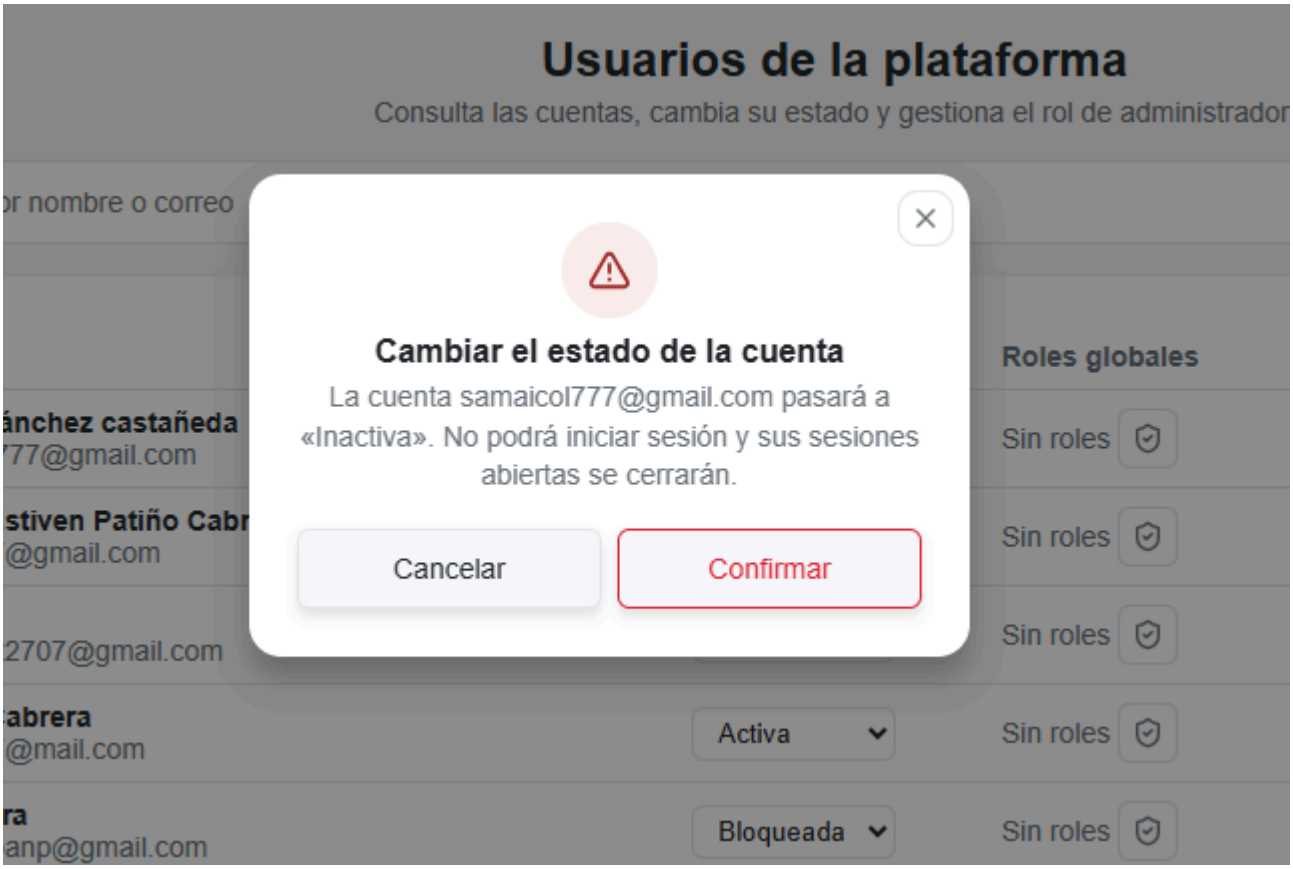
Field	Detail
Role that can access	Administrator with the account management permission.
Access path	Platform users > Status column of the account.
Screen components	Selector with Active, Inactive and Blocked; confirmation window.

Steps to use it:

> 1. On the account's row, choose the new status.
> 2. If the new status is Inactive or Blocked, the application warns: "The account will become … It will not be able to sign in and its open sessions will be closed.". Press Confirm.

Expected result: The status changes. An inactive or blocked account cannot sign in; going back to Active asks for no confirmation.

Notes: The session the account already had open may last up to 15 minutes, but it is not renewed.



"Change the account status" confirmation (web)

15.4 Grant or revoke the administrator role

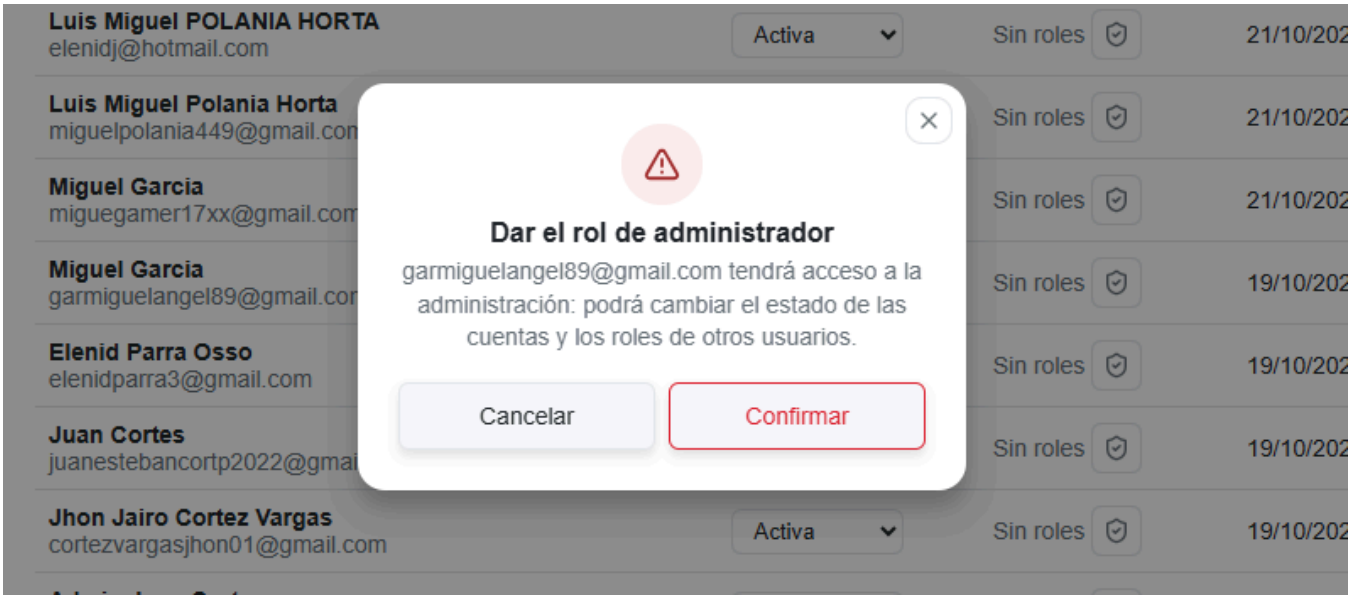
Description: Makes an account an administrator or removes the role.

Field	Detail
Role that can access	Administrator with the role management permission.
Access path	Platform users > shield button in the Global roles column.
Screen components	"Make … administrator" or "Remove the administrator role from …" button; confirmation window.

Steps to use it:

1. Press the shield on the account's row.
2. Read the confirmation ("Grant the administrator role": they will be able to change the status of accounts and the roles of other users; or "Remove the administrator role") and press Confirm.

Expected result: The role appears or disappears in the Global roles column. The account must sign out and in again to see the change in its menu.



"Grant the administrator role" confirmation (web)

15.5 Administration notifications

Description: Notices about the platform for the administrators.

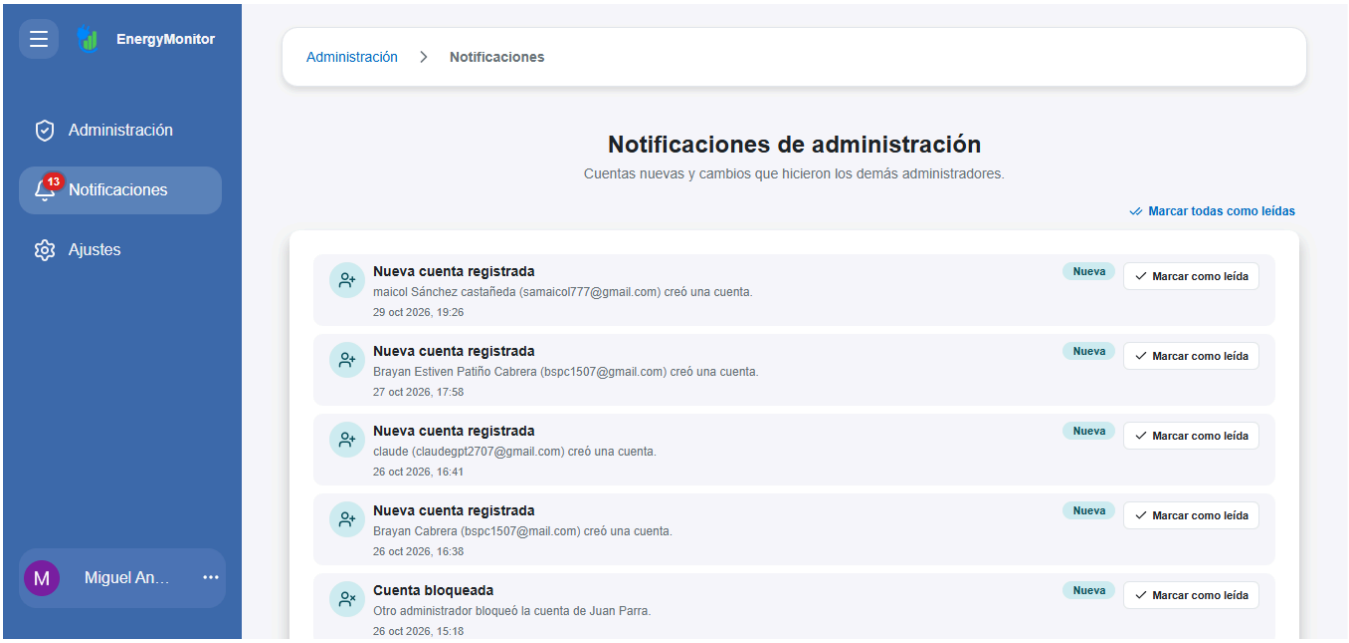
Field	Detail
Role that can access	Administrator.
Access path	Side menu > Notifications.
Screen components	"Administration notifications" title; Mark all as read; cards with title, message, status (New or Read) and the Mark as read and Delete buttons.

Notice	When it arrives
New account registered	Someone creates an account (with email or with their first Google sign-in).
Account activated / deactivated / blocked	Another administrator changes the status of an account.
New administrator / Administrator removed	Another administrator grants or revokes the administrator role.

Steps to use it: open Notifications, read the notices, mark them as read or delete them.

Expected result: The administrator keeps up with new accounts and with the changes made by the other administrators.

Notes: The 50 most recent notices are shown. Whoever makes a change gets no notice of their own change. These notices are only seen on the web; they do not arrive by email or push.

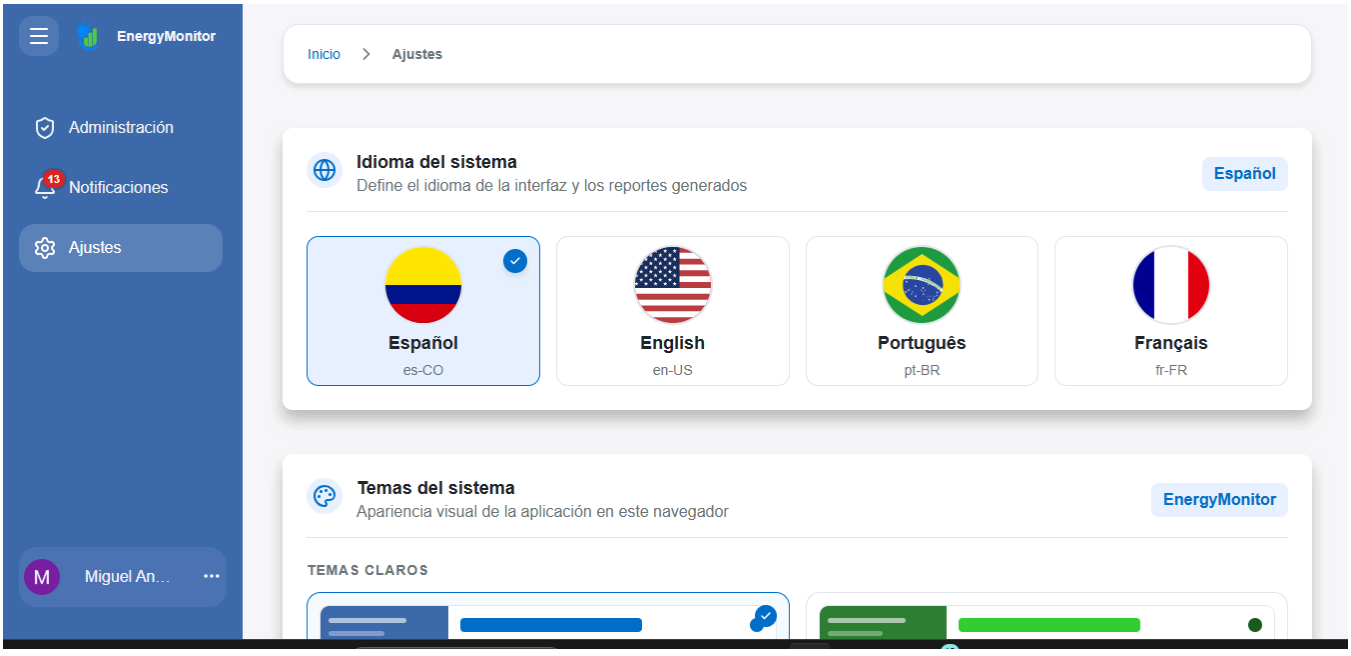


Administration notifications (web)

15.6 Administrator settings and restrictions

In **Settings**, the administrator sees the language, the theme and the help center, but not the notification preferences of the homes. Also:

- An administrator cannot change their own status or roles: their row shows no actions; another administrator must do it.
- An action without the matching permission is not shown.
- The administrator role gives no access to the homes.



Settings of an administrator account, without the notifications section (web)

System messages

System message	Meaning	User action
Required field / invalid format	A piece of data is missing or does not have the expected format.	Fix the marked field.
Invalid user or password	The credentials do not match.	Check the data or recover the password.
Email already registered	The email already has an account.	Sign in or recover the password.
The email domain looks mistyped / does not exist	The email has a typo or a domain without email.	Fix the email (the system suggests the likely provider).
You must accept the terms	The acceptance of the terms and privacy policy was not ticked.	Read them and tick the checkbox.
Verify your email to sign in	The account exists but the email has not been verified.	Type the code that arrives at the email.
Invalid or expired verification code	The code does not match or has expired.	Ask for a new code.
Too many attempts	The allowed number of attempts was exceeded.	Wait a few minutes and try again.
Email verified. You can now sign in.	The account was created.	Sign in.
Password updated	The new password was saved.	Sign in with the new password.
The code must have 8 characters (letters and digits)	The home access code is incomplete.	Type the full code the owner shared.
The network password is not correct / The module cannot find that network	The module's Wi-Fi connection failed while linking.	Check the password or move the module closer to the router.
The Bluetooth connection with the module was lost	The module moved away or turned off during linking.	Move closer and retry.
This browser cannot connect over Bluetooth	The browser does not support Web Bluetooth.	Use Chrome or Edge, the mobile app or the "EnergyMonitor-Setup" portal.
Only the owner of the home can change these thresholds	A member tries to edit the thresholds.	Ask the owner for the change.
The daily limit cannot exceed the monthly one	Inconsistent threshold values.	Fix the values.
The action could not be completed. Try again.	An action failed because of a temporary problem.	Retry; if it persists, contact support.

Frequently asked questions

Can I use the same account on the web and on the phone?

Yes. The account, homes, devices and notifications are the same in both applications.

How do I invite someone to my home?

By sharing the 8-character access code shown in the Home tab (only the owner sees it). The other person creates their account and uses Join home. There are no email invitations.

What do I do if I forgot my password?

On Sign in, press Forgot your password?; a code arrives at the email to set a new one.

What if the verification code does not arrive?

Check the spam folder. If it does not show up, ask for it again from the same screen.

I registered with Google, how do I delete my account?

In My account > Delete account, press Send me the code and type the code that arrives at the Google email.

Can I stop being the owner of a home?

Yes: in the Users tab, "Make owner" another member. Afterwards you can leave the home if you wish.

Why can't I link a device from my browser?

Browser Bluetooth linking only works in Chrome or Edge on a computer and in Chrome on Android. Use the mobile app or the "EnergyMonitor-Setup" portal.

When does a recommendation appear?

The system reviews the consumption of each home every hour; recommendations appear when a pattern is detected (for example, consumption at night) and need at least a few days of readings.

Can I change the language or the theme whenever I want?

Yes, from Settings, as many times as you want.

Is there an iPhone app?

No. On iPhone, use the web application.

Does EnergyMonitor work without internet?

No. The applications and the modules need a connection to send and show the data.

Glossary

Term	Description
ESP32	Microcontroller of the measuring module; sends the readings over Wi-Fi and is configured over Bluetooth.
PZEM-004T V3 sensor	Sensor that measures voltage, current, power and energy of the appliance.
kWh	Kilowatt-hour, unit of electrical energy consumption.
Home	House, apartment or other place monitored in EnergyMonitor.
Owner	Responsible for a home; manages members, devices and thresholds.
Member	Person who joined a home with the access code; views its information.
Access code	8-character code other people use to join a home.
Verification code	6-digit code sent by email to verify the account, recover the password or delete a Google account.
Threshold	Daily or monthly consumption limit of a home that triggers alerts.
Alert	Notice about something that needs attention (high consumption, limit reached, device disconnected).
Recommendation	Saving suggestion generated from consumption.
Activity	Notice of what happened in the account, its homes or its devices.
Push notification	Notice that reaches the browser or the phone even when the application is closed.
Administrator	Account that manages the platform accounts from the web.
Session	Period in which the account stays authenticated, from signing in until signing out.

Annexes

Annex A. Images of the manual. The screenshots are stored in `12-manual-usuario/assets/Web` (web application) and `12-manual-usuario/assets/Mobile` (mobile application). They follow the naming convention of the export tool: `image*.png` in the web folder and `image*.png` or `image*.jpeg` in the mobile folder. Every image referenced in the text is included in those folders and corresponds to the current version of each application; the full index is the file listing of both folders.

Annex B. Sources. This manual was written from the code of the web application (Energy-Monitor-application) and of the mobile application (Energy-Monitor-Mobile) as of October 27, 2026, from their interface texts and from the requirements specification, following the guide for writing software manuals of the ADSO program. It replaces the separate "EnergyMonitor Web user manual" and "EnergyMonitor Mobile user manual" (version 2.0).